

Higgs Boson at Large Hadron Collider

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Standard Model

▷ Gauge theory of interactions between fermions mediated by vector bosons

- Electroweak interaction: $SU(2) \times U(1)$
- Quantum chromodynamics (strong) interaction: $SU(3)$
- No gravitation

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + i\bar{\psi} \not{D} \psi + h.c.$$

▷ Electroweak Lagrangian $\mathcal{L}_{\text{symm}} = -\frac{1}{4} \sum_{A=1}^3 F_{\mu\nu}^A F^{A\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} + \bar{\psi}_L i\gamma^\mu D_\mu \psi_L + \bar{\psi}_R i\gamma^\mu D_\mu \psi_R .$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$$

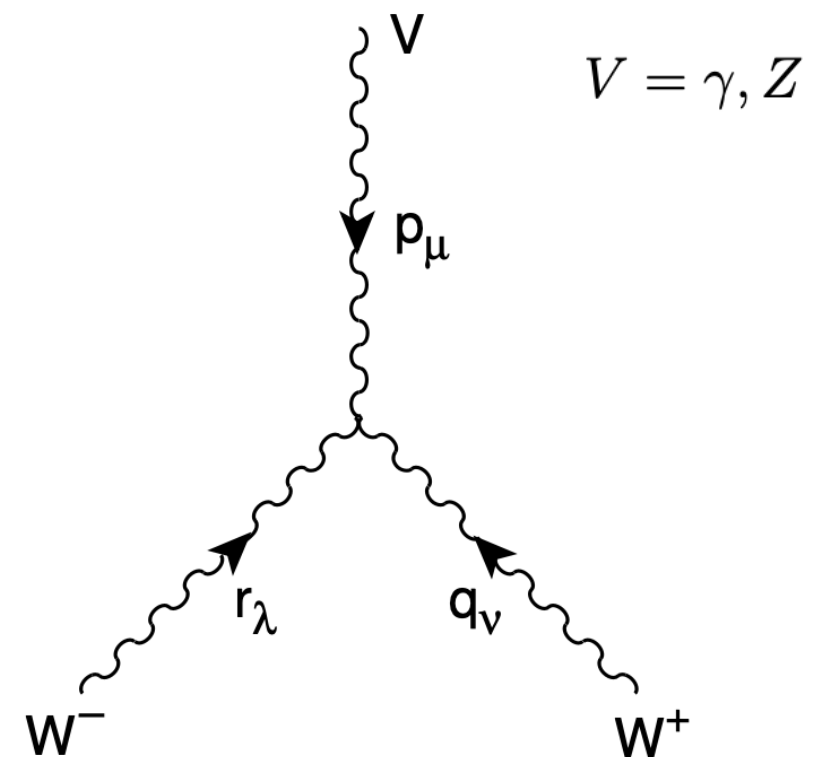
$$F_{\mu\nu}^A = \partial_\mu W_\nu^A - \partial_\nu W_\mu^A - g\epsilon_{ABC} W_\mu^B W_\nu^C$$

$$\psi_{L,R} = [(1 \mp \gamma_5)/2]\psi, \quad \bar{\psi}_{L,R} = \bar{\psi}[(1 \pm \gamma_5)/2]$$

$$D_\mu \psi_{L,R} = \left[\partial_\mu + ig \sum_{A=1}^3 t_{L,R}^A W_\mu^A + ig' \frac{1}{2} Y_{L,R} B_\mu \right] \psi_{L,R}$$

$$A_\mu = \cos \theta_W B_\mu + \sin \theta_W W_\mu^3 ,$$

$$Z_\mu = -\sin \theta_W B_\mu + \cos \theta_W W_\mu^3 .$$



$$g_{W^-W^+\gamma} = g \sin \theta_W = e \quad \text{and} \quad g_{W^-W^+Z} = g \cos \theta_W$$

The Problem and A Solution

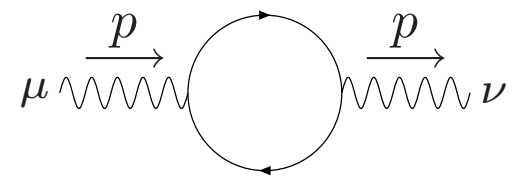
- ▷ Quantum field theory of fundamental forces and matter works fine *IF* all particles are massless
- ▷ No mass-term for fermions $-m\bar{\psi}\psi = -m(\bar{\psi}_R\psi_L + \bar{\psi}_L\psi_R)$
 - Such a term mixes left-handed and right-handed spinors and is inconsistent with the chiral SU(2) symmetry
- ▷ We have experimental evidence that, except for photons, all known particles are massive
 - Massless particles: $E = pc$
 - Massive particles: $E = \sqrt{(mc^2)^2 + (pc)^2}$

A new scalar elementary particle

- ▷ Discovery of W and Z in 1984 proved them to be pretty heavy
- ▷ Need a mechanism to dynamically generate mass for all particles and conserve the gauge invariance
- ▷ Higgs mechanism and goldstone bosons address this problem
- ▷ Experimental evidence needed to validate theory
 - Predict Higgs production processes and decay rates
- ▷ Direct search
 - produce Higgs and look for decay products: invariant mass and characteristic kinematics
- ▷ Indirect search
 - Calculate contribution of Higgs in corrections to well known SM processes
 - Combined analysis of all precisely measured Z-pole observables and check validity with different Higgs mass assumptions

Generating Mass for W and Z

- ▷ Propagator for massless vector bosons at tree level



$$i \Pi_{\mu\nu}(p) \equiv i(p_\mu p_\nu - p^2 g_{\mu\nu}) \Pi(p^2)$$

Gauge invariance

$$p^\mu \Pi_{\mu\nu}(p) = p^\nu \Pi_{\mu\nu}(p) = 0$$

- ▷ Renormalized propagator at higher order

$$\text{wavy line} + \text{wavy line with loop} + \text{wavy line with two loops} + \dots = \frac{-i(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2})}{p^2 [1 + \Pi(p^2)]}$$

- ▷ For a massless scalar particle in the loop

$$\text{wavy line with loop} \quad \Pi(p^2) \underset{p^2 \rightarrow 0}{\simeq} \frac{-g^2 v^2}{p^2} \longrightarrow p^2 [1 + \Pi(p^2)] = p^2 - g^2 v^2$$

- ▷ For $p=0$ we have our vector boson with $m = gv$!

- ▷ Do we have already any such scalar?

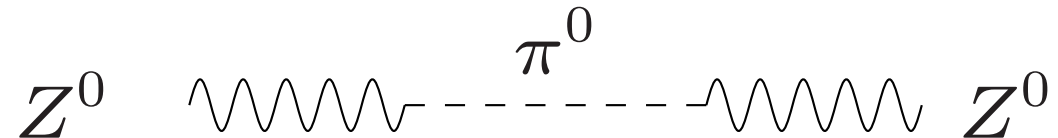
Generating W and Z Mass with π^0

- ▷ W and Z couple to charge and neutral currents

$$\mathcal{L}_{\text{int}} = g_Z j_\mu^Z Z^\mu + g_W (j_\mu^W W^{+\mu} + \text{h.c.})$$

- ▷ Charged and neutral pions have very small mass (compared to other hadrons) and are massless in the limit of massless quarks!

- ▷ Pions in the loop provided needed mass term $\Pi(p^2) = -g_Z^2 f_\pi^2 / p^2$



The diagram shows a Z⁰ boson (represented by a wavy line) entering from the left, interacting with a loop containing a pi⁰ meson (represented by a dashed line), and then exiting to the right as another Z⁰ boson (wavy line).

- ▷ Prediction for W and Z mass $m_W = g_W f_\pi$ $m_Z = g_Z f_\pi$

- ▷ Pion decay constant measured from decay rate to be $f_\pi = 93$ MeV

- ▷ W and Z mass related in SM through $\frac{m_W}{m_Z} = \frac{g_W}{g_Z} \equiv \cos \theta_W \simeq 0.88$
 - Good agreement with measurement

- ▷ We know $g_Z \simeq 0.37$ which corresponds to $m_Z = 35$ MeV!
 - Off by 3 orders of magnitude!

- ▷ Need a new scalar for dynamic mass generation

Spontaneous Symmetry Breaking

- ▷ New complex SU(2) doublet with four degrees of freedom (massless scalar particles)

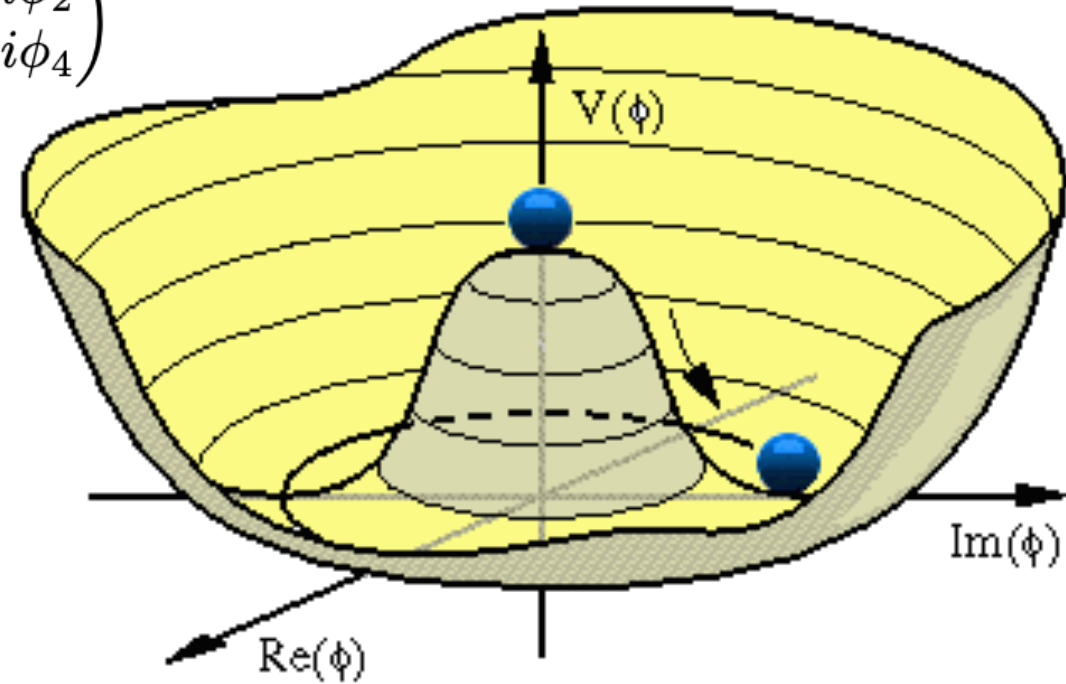
$$\phi \equiv \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix}$$

- ▷ Self-interaction potential

$$V(\Phi) = \frac{\lambda}{4} (\Phi^\dagger \Phi - \frac{1}{2}v^2)^2$$

- ▷ Non-zero vacuum expectation value (VEV) generates one massive scalar

- remaining 3 scalar absorbed by W, Z, and photon



$$\langle \Phi^0 \rangle = v/\sqrt{2}$$

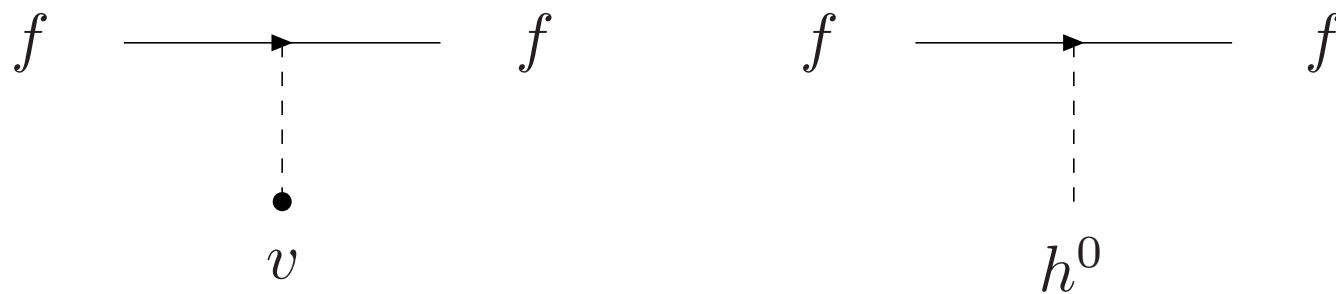
- ▷ We want this scalar to couple to generate correct mass $m_Z = g_Z v \simeq 91 \text{ GeV}$
- ▷ Therefore the VEV must be $v = 246 \text{ GeV}$

- ▷ New massive scalar particle commonly known as Higgs Boson $m_h = \frac{1}{2}\lambda v^2$
 - Interactions, decay rates known precisely
 - But its **mass not fixed and must be measured**

Fermion Mass thanks to Higgs

- ▷ In addition to Z and W, also fermions acquire mass in SM thanks to Yukawa coupling to Higgs boson

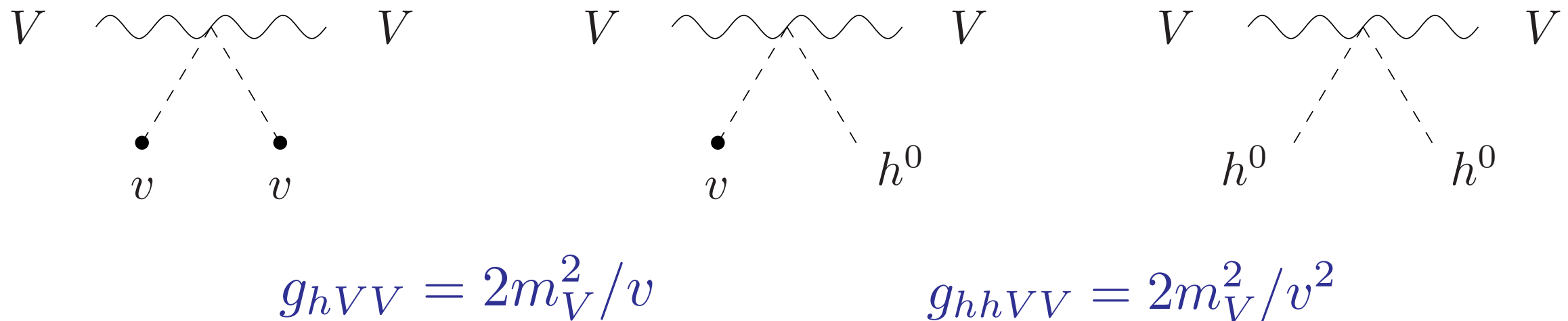
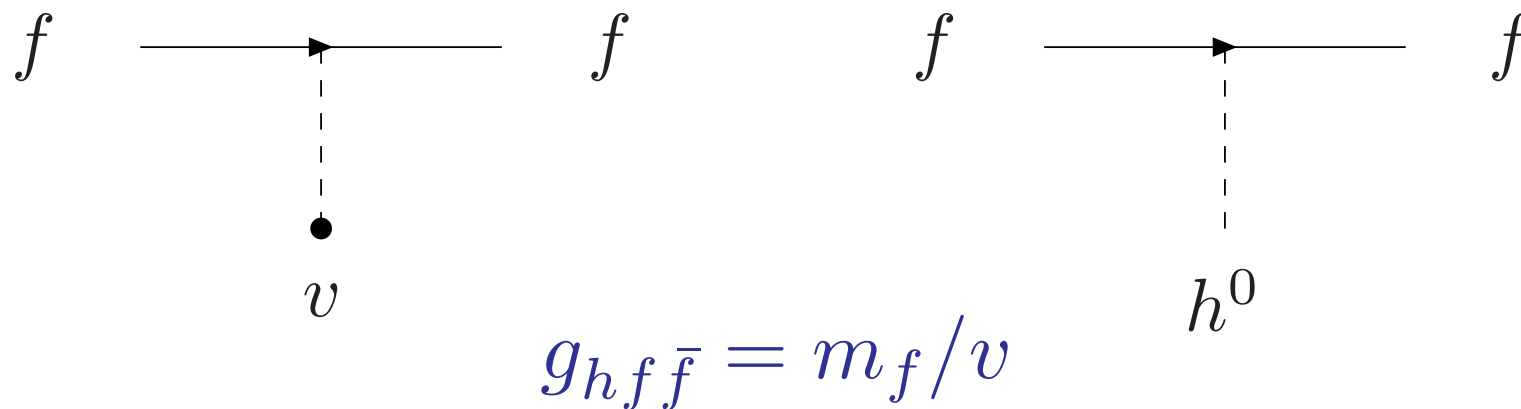
$$\mathcal{L}_{\text{Yukawa}} = -h_u(\bar{u}_R u_L \Phi^0 - \bar{u}_R d_L \Phi^+) - h_d(\bar{d}_R d_L \Phi^{0*} + \bar{d}_R u_L \Phi^-) + \text{h.c.}$$



fermions	SU(2)	U(1) _Y
(ν , e^-) _L	2	-1
e_R^-	1	-2
(u , d) _L	2	1/3
u_R	1	4/3
d_R	1	-2/3

- ▷ Coupling proportional to fermion mass $g_{h f \bar{f}} = m_f / v$
 - Will determine Higgs decay branching fractions

Higgs Coupling



- ▷ Higgs coupling enhanced for heavier particles
- ▷ Vector bosons always preferred to fermions
 - But must be kinematically allowed
- ▷ Fixing mass of Higgs fixes decay rates for all final states

The complete Lagrangian

QCD

Electroweak + Higgs

Fermion interaction
with weak and Higgs

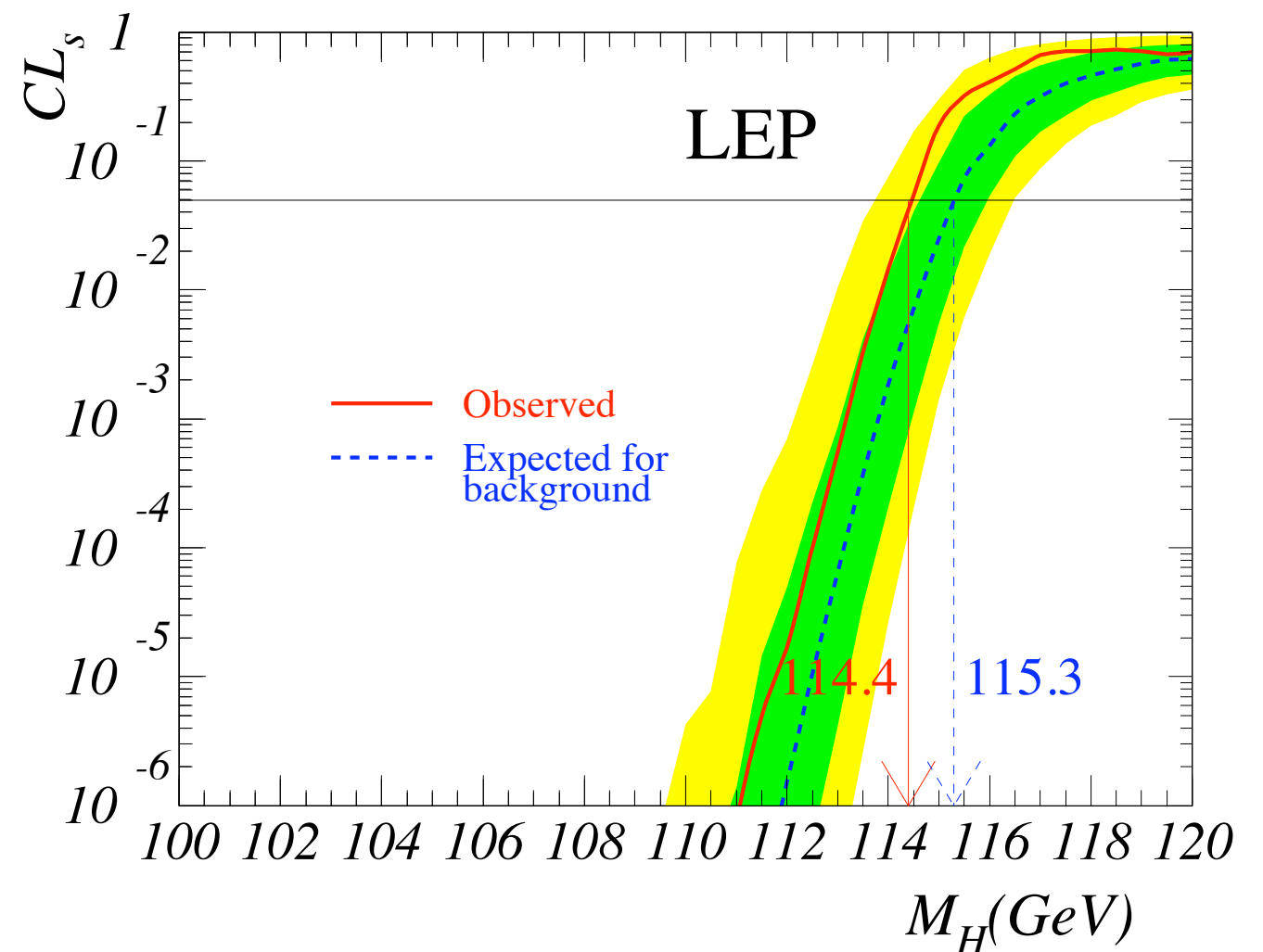
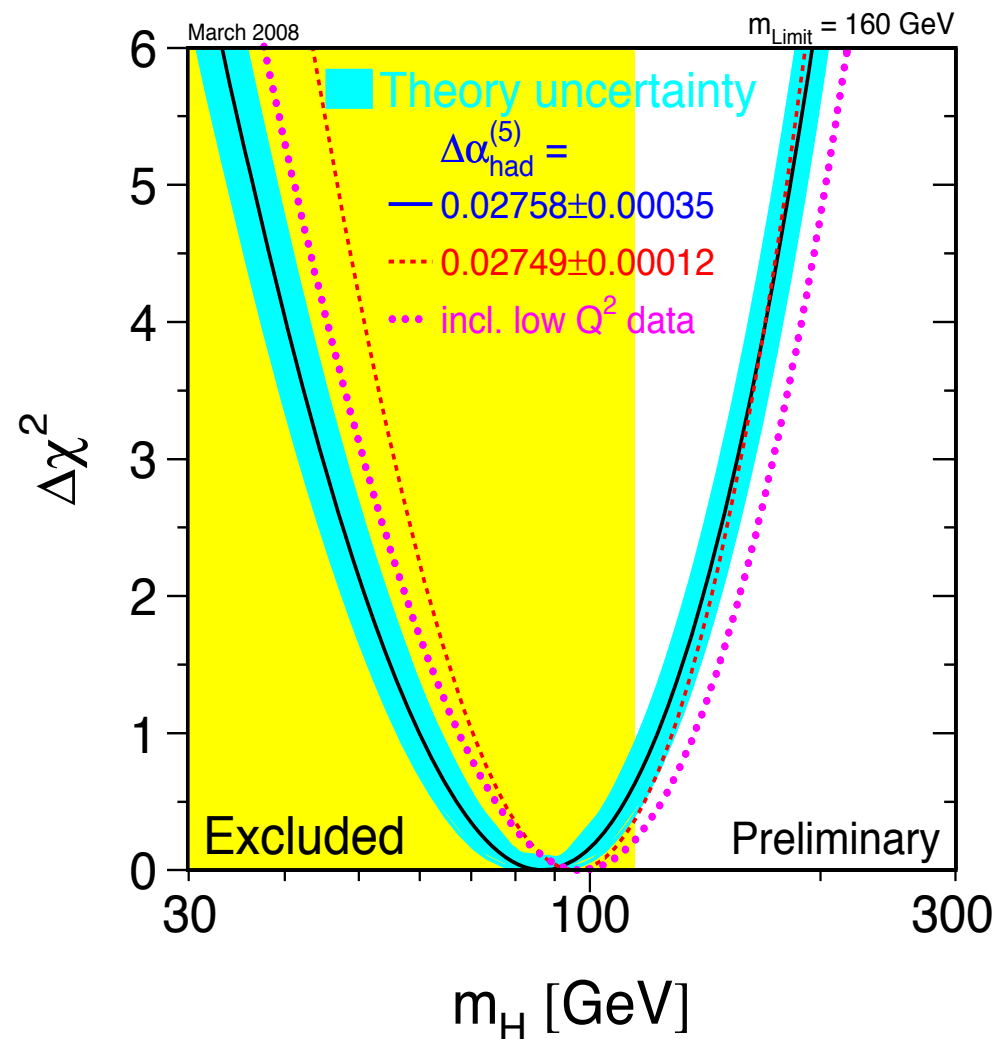
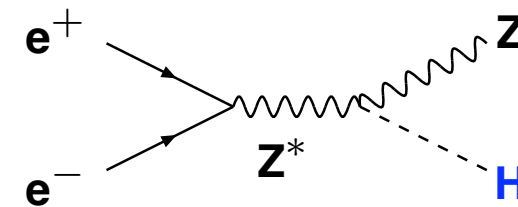
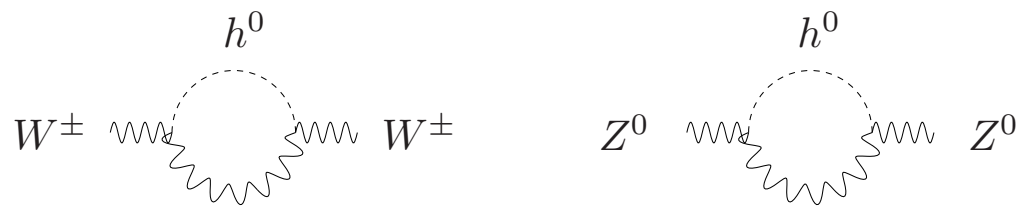
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$$\begin{aligned}
 & 1 \quad -\frac{1}{2}\partial_\nu g_\mu^a \partial_\nu g_\mu^a - g_s f^{abc} \partial_\mu g_\nu^a g_\mu^b g_\nu^c - \frac{1}{4}g_s^2 f^{abc} f^{ade} g_\mu^b g_\nu^c g_\mu^d g_\nu^e + \\
 & \quad \frac{1}{2}ig_s^2(\bar{q}_i^\sigma \gamma^\mu q_j^\sigma)g_\mu^a + \bar{G}^a \partial^2 G^a + g_s f^{abc} \partial_\mu \bar{G}^a G^b g_\mu^c - \partial_\nu W_\mu^+ \partial_\nu W_\mu^- - \\
 & 2 \quad M^2 W_\mu^+ W_\mu^- - \frac{1}{2}\partial_\nu Z_\mu^0 \partial_\nu Z_\mu^0 - \frac{1}{2c_w^2}M^2 Z_\mu^0 Z_\mu^0 - \frac{1}{2}\partial_\mu A_\nu \partial_\mu A_\nu - \frac{1}{2}\partial_\mu H \partial_\mu H - \\
 & \quad \frac{1}{2}m_h^2 H^2 - \partial_\mu \phi^+ \partial_\mu \phi^- - M^2 \phi^+ \phi^- - \frac{1}{2}\partial_\mu \phi^0 \partial_\mu \phi^0 - \frac{1}{2c_w^2}M\phi^0 \phi^0 - \beta_h \left[\frac{2M^2}{g^2} + \right. \\
 & \quad \left. \frac{2M}{g}H + \frac{1}{2}(H^2 + \phi^0 \phi^0 + 2\phi^+ \phi^-) \right] + \frac{2M^4}{g^2}\alpha_h - igc_w[\partial_\nu Z_\mu^0(W_\mu^+ W_\nu^- - \\
 & \quad W_\nu^+ W_\mu^-) - Z_\nu^0(W_\mu^+ \partial_\nu W_\mu^- - W_\mu^- \partial_\nu W_\mu^+) + Z_\mu^0(W_\nu^+ \partial_\nu W_\mu^- - \\
 & \quad W_\nu^- \partial_\nu W_\mu^+)] - ig s_w[\partial_\nu A_\mu(W_\mu^+ W_\nu^- - W_\nu^+ W_\mu^-) - A_\nu(W_\mu^+ \partial_\nu W_\mu^- - \\
 & \quad W_\mu^- \partial_\nu W_\mu^+) + A_\mu(W_\nu^+ \partial_\nu W_\mu^- - W_\nu^- \partial_\nu W_\mu^+)] - \frac{1}{2}g^2 W_\mu^+ W_\mu^- W_\nu^+ W_\nu^- + \\
 & \quad \frac{1}{2}g^2 W_\mu^+ W_\nu^- W_\mu^- W_\nu^+ + g^2 c_w^2(Z_\mu^0 W_\mu^+ Z_\nu^0 W_\nu^- - Z_\mu^0 Z_\nu^0 W_\mu^+ W_\nu^-) + \\
 & \quad g^2 s_w^2(A_\mu W_\mu^+ A_\nu W_\nu^- - A_\mu A_\nu W_\mu^+ W_\nu^-) + g^2 s_w c_w[A_\mu Z_\nu^0(W_\mu^+ W_\nu^- - \\
 & \quad W_\nu^+ W_\mu^-) - 2A_\mu Z_\mu^0 W_\nu^+ W_\nu^-] - g\alpha[H^3 + H\phi^0 \phi^0 + 2H\phi^+ \phi^-] - \\
 & \quad \frac{1}{8}g^2 \alpha_h[H^4 + (\phi^0)^4 + 4(\phi^+ \phi^-)^2 + 4(\phi^0)^2 \phi^+ \phi^- + 4H^2 \phi^+ \phi^- + 2(\phi^0)^2 H^2] - \\
 & \quad gM W_\mu^+ W_\mu^- H - \frac{1}{2}g\frac{M}{c_w^2}Z_\mu^0 Z_\mu^0 H - \frac{1}{2}ig[W_\mu^+(\phi^0 \partial_\mu \phi^- - \phi^- \partial_\mu \phi^0) - \\
 & \quad W_\mu^-(\phi^0 \partial_\mu \phi^+ - \phi^+ \partial_\mu \phi^0)] + \frac{1}{2}g[W_\mu^+(H \partial_\mu \phi^- - \phi^- \partial_\mu H) - W_\mu^-(H \partial_\mu \phi^+ - \\
 & \quad \phi^+ \partial_\mu H)] + \frac{1}{2}g\frac{1}{c_w}(Z_\mu^0(H \partial_\mu \phi^0 - \phi^0 \partial_\mu H) - ig\frac{s_w^2}{c_w}M Z_\mu^0(W_\mu^+ \phi^- - W_\mu^- \phi^+) + \\
 & \quad ig s_w M A_\mu(W_\mu^+ \phi^- - W_\mu^- \phi^+) - ig\frac{1-2c_w^2}{2c_w}Z_\mu^0(\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) + \\
 & \quad ig s_w A_\mu(\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) - \frac{1}{4}g^2 W_\mu^+ W_\mu^- [H^2 + (\phi^0)^2 + 2\phi^+ \phi^-] - \\
 & \quad \frac{1}{4}g^2 \frac{1}{c_w^2}Z_\mu^0 Z_\mu^0 [H^2 + (\phi^0)^2 + 2(2s_w^2 - 1)\phi^+ \phi^-] - \frac{1}{2}g^2 \frac{s_w^2}{c_w}Z_\mu^0 \phi^0(W_\mu^+ \phi^- + \\
 & \quad W_\mu^- \phi^+) - \frac{1}{2}ig^2 \frac{s_w^2}{c_w}Z_\mu^0 H(W_\mu^+ \phi^- - W_\mu^- \phi^+) + \frac{1}{2}g^2 s_w A_\mu \phi^0(W_\mu^+ \phi^- + \\
 & \quad W_\mu^- \phi^+) + \frac{1}{2}ig^2 s_w A_\mu H(W_\mu^+ \phi^- - W_\mu^- \phi^+) - g^2 \frac{s_w}{c_w}(2c_w^2 - 1)Z_\mu^0 A_\mu \phi^+ \phi^- - \\
 & \quad g^1 s_w^2 A_\mu A_\mu \phi^+ \phi^- - \bar{e}^\lambda (\gamma \partial + m_e^\lambda) e^\lambda - \bar{\nu}^\lambda \gamma \partial \nu^\lambda - \bar{u}_j^\lambda (\gamma \partial + m_u^\lambda) u_j^\lambda - \\
 & 3 \quad \bar{d}_j^\lambda (\gamma \partial + m_d^\lambda) d_j^\lambda + ig s_w A_\mu [-(\bar{e}^\lambda \gamma^\mu e^\lambda) + \frac{2}{3}(\bar{u}_j^\lambda \gamma^\mu u_j^\lambda) - \frac{1}{3}(\bar{d}_j^\lambda \gamma^\mu d_j^\lambda)] + \\
 & \quad \frac{ig}{4c_w}Z_\mu^0[(\bar{\nu}^\lambda \gamma^\mu (1 + \gamma^5) \nu^\lambda) + (\bar{e}^\lambda \gamma^\mu (4s_w^2 - 1 - \gamma^5) e^\lambda) + (\bar{u}_j^\lambda \gamma^\mu (\frac{4}{3}s_w^2 - \\
 & \quad 1 - \gamma^5) u_j^\lambda) + (\bar{d}_j^\lambda \gamma^\mu (1 - \frac{8}{3}s_w^2 - \gamma^5) d_j^\lambda)] + \frac{ig}{2\sqrt{2}}W_\mu^+[(\bar{\nu}^\lambda \gamma^\mu (1 + \gamma^5) e^\lambda) + \\
 & \quad (\bar{u}_j^\lambda \gamma^\mu (1 + \gamma^5) C_{\lambda\kappa} d_j^\kappa)] + \frac{ig}{2\sqrt{2}}W_\mu^-[(\bar{e}^\lambda \gamma^\mu (1 + \gamma^5) \nu^\lambda) + (\bar{d}_j^\kappa C_{\lambda\kappa}^\dagger \gamma^\mu (1 + \\
 & \quad \gamma^5) u_j^\lambda)] + \frac{ig}{2\sqrt{2}}\frac{m_e^\lambda}{M}[-\phi^+(\bar{\nu}^\lambda (1 - \gamma^5) e^\lambda) + \phi^-(\bar{e}^\lambda (1 + \gamma^5) \nu^\lambda)] - \\
 & 4 \quad \frac{g}{2}\frac{m_e^\lambda}{M}[H(\bar{e}^\lambda e^\lambda) + i\phi^0(\bar{e}^\lambda \gamma^5 e^\lambda)] + \frac{ig}{2M\sqrt{2}}\phi^+[-m_d^\kappa(\bar{u}_j^\lambda C_{\lambda\kappa}(1 - \gamma^5) d_j^\kappa) + \\
 & \quad m_u^\lambda(\bar{u}_j^\lambda C_{\lambda\kappa}(1 + \gamma^5) d_j^\kappa) + \frac{ig}{2M\sqrt{2}}\phi^-[m_d^\lambda(\bar{d}_j^\lambda C_{\lambda\kappa}^\dagger(1 + \gamma^5) u_j^\kappa) - m_u^\kappa(\bar{d}_j^\lambda C_{\lambda\kappa}^\dagger(1 - \\
 & \quad \gamma^5) u_j^\kappa) - \frac{g}{2}\frac{m_u^\lambda}{M}H(\bar{u}_j^\lambda u_j^\lambda) - \frac{g}{2}\frac{m_d^\lambda}{M}H(\bar{d}_j^\lambda d_j^\lambda) + \frac{ig}{2}\frac{m_u^\lambda}{M}\phi^0(\bar{u}_j^\lambda \gamma^5 u_j^\lambda) - \\
 & \quad \frac{ig}{2}\frac{m_d^\lambda}{M}\phi^0(\bar{d}_j^\lambda \gamma^5 d_j^\lambda) + \bar{X}^+(\partial^2 - M^2)X^+ + \bar{X}^-(\partial^2 - M^2)X^- + \bar{X}^0(\partial^2 - \\
 & 5 \quad \frac{M^2}{c_w^2})X^0 + \bar{Y}\partial^2 Y + igc_w W_\mu^+(\partial_\mu \bar{X}^0 X^- - \partial_\mu \bar{X}^+ X^0) + ig s_w W_\mu^+(\partial_\mu \bar{Y} X^- - \\
 & \quad \partial_\mu \bar{X}^+ Y) + igc_w W_\mu^-(\partial_\mu \bar{X}^- X^0 - \partial_\mu \bar{X}^0 X^+) + ig s_w W_\mu^-(\partial_\mu \bar{X}^- Y - \\
 & \quad \partial_\mu \bar{Y} X^+) + igc_w Z_\mu^0(\partial_\mu \bar{X}^+ X^- - \partial_\mu \bar{X}^- X^+) + ig s_w A_\mu(\partial_\mu \bar{X}^+ X^- - \\
 & \quad \partial_\mu \bar{X}^- X^+) - \frac{1}{2}gM[\bar{X}^+ X^+ H + \bar{X}^- X^- H + \frac{1}{c_w^2}\bar{X}^0 X^0 H] + \\
 & \quad \frac{1-2c_w^2}{2c_w}igM[\bar{X}^+ X^0 \phi^+ - \bar{X}^- X^0 \phi^-] + \frac{1}{2c_w}igM[\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \\
 & \quad igM s_w[\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \frac{1}{2}igM[\bar{X}^+ X^+ \phi^0 - \bar{X}^- X^- \phi^0]
 \end{aligned}$$

Experimental Constraints on Higgs Mass

- ▷ Indirect search through precision measurement of SM sensitive to radiative corrections to Higgs in loops

- Direct search at LEP in Higgs-strahlung production channel

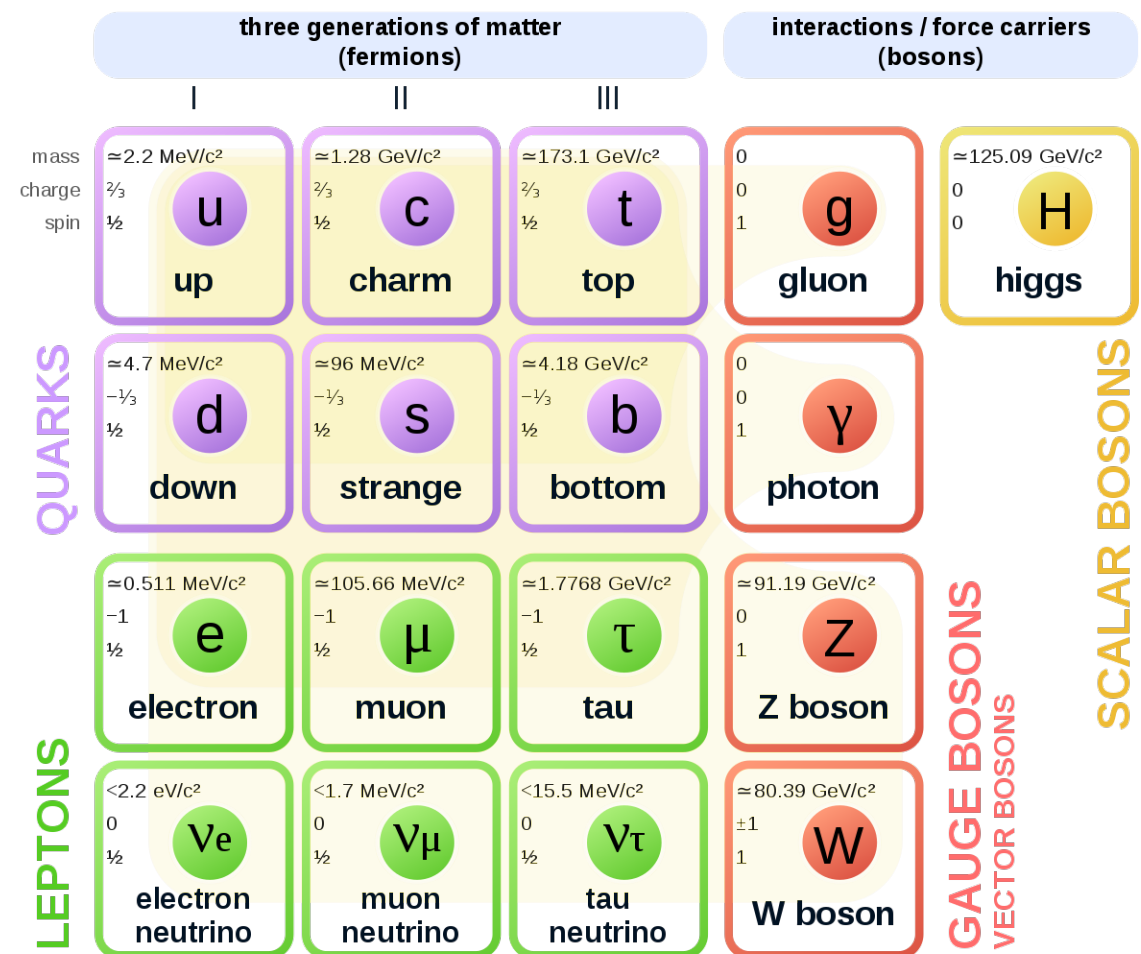


The Standard Model Lagrangian

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\ & + i \bar{\psi} \not{D} \psi + \text{h.c.} \\ & + \bar{\psi}_i Y_{ij} \psi_j \phi + \text{h.c.} \\ & + |D_\mu \phi|^2 - V(\phi)\end{aligned}$$

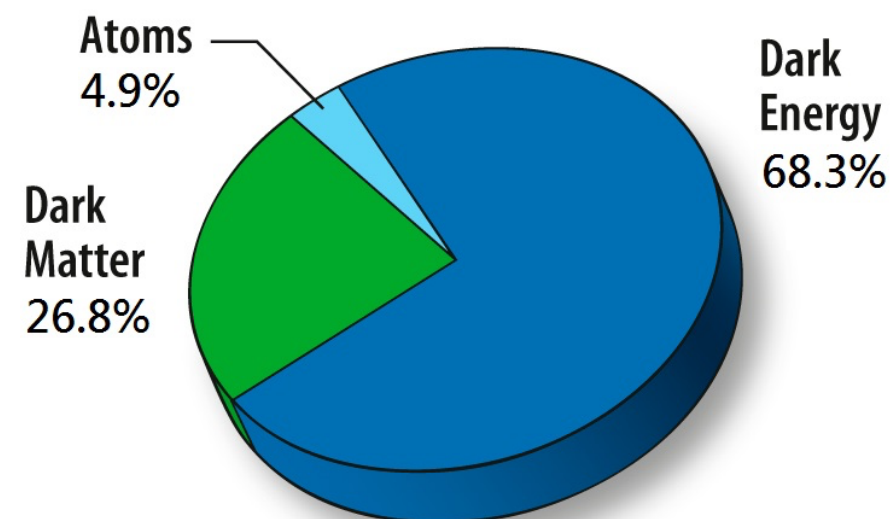
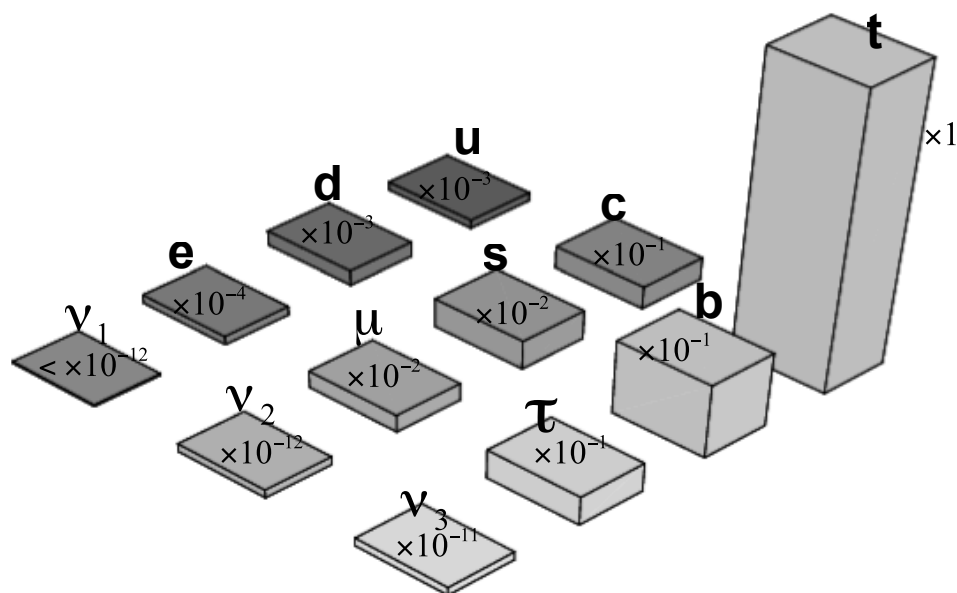
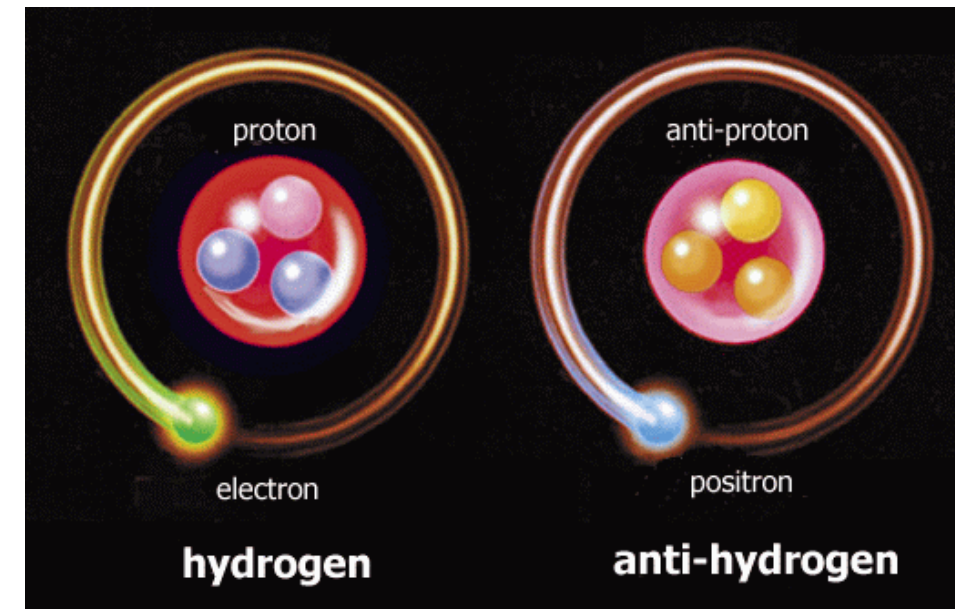
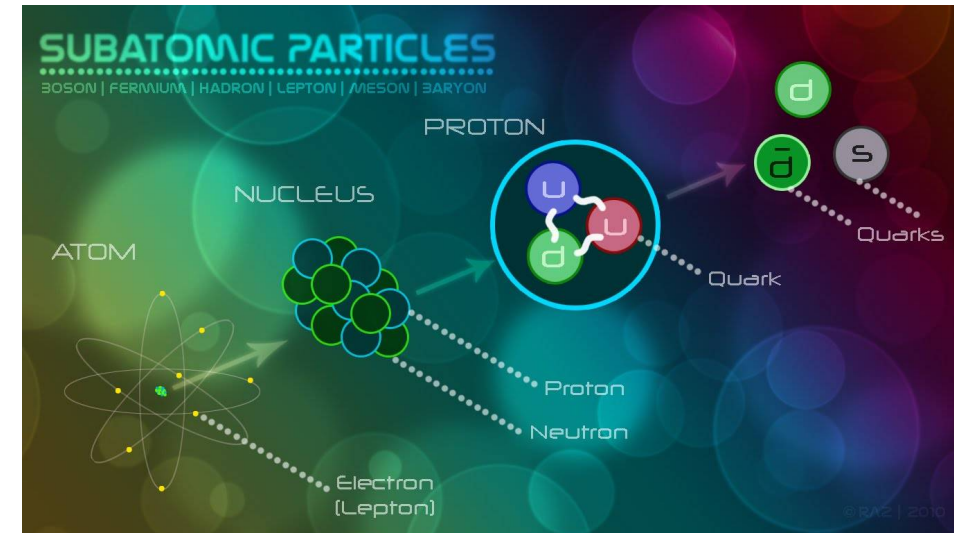
Testing the Standard Model

- ▷ Extremely predictive theory since its inception
- ▷ Last missing piece discovered just 13 years ago
 - Compare to gravitational waves and general relativity
- ▷ Has successfully resisted 50 years of falsification
- ▷ *We already know it is incomplete*
 - e.g. neutrinos are massive
- ▷ It cannot address some basic curiosities and questions about our Universe

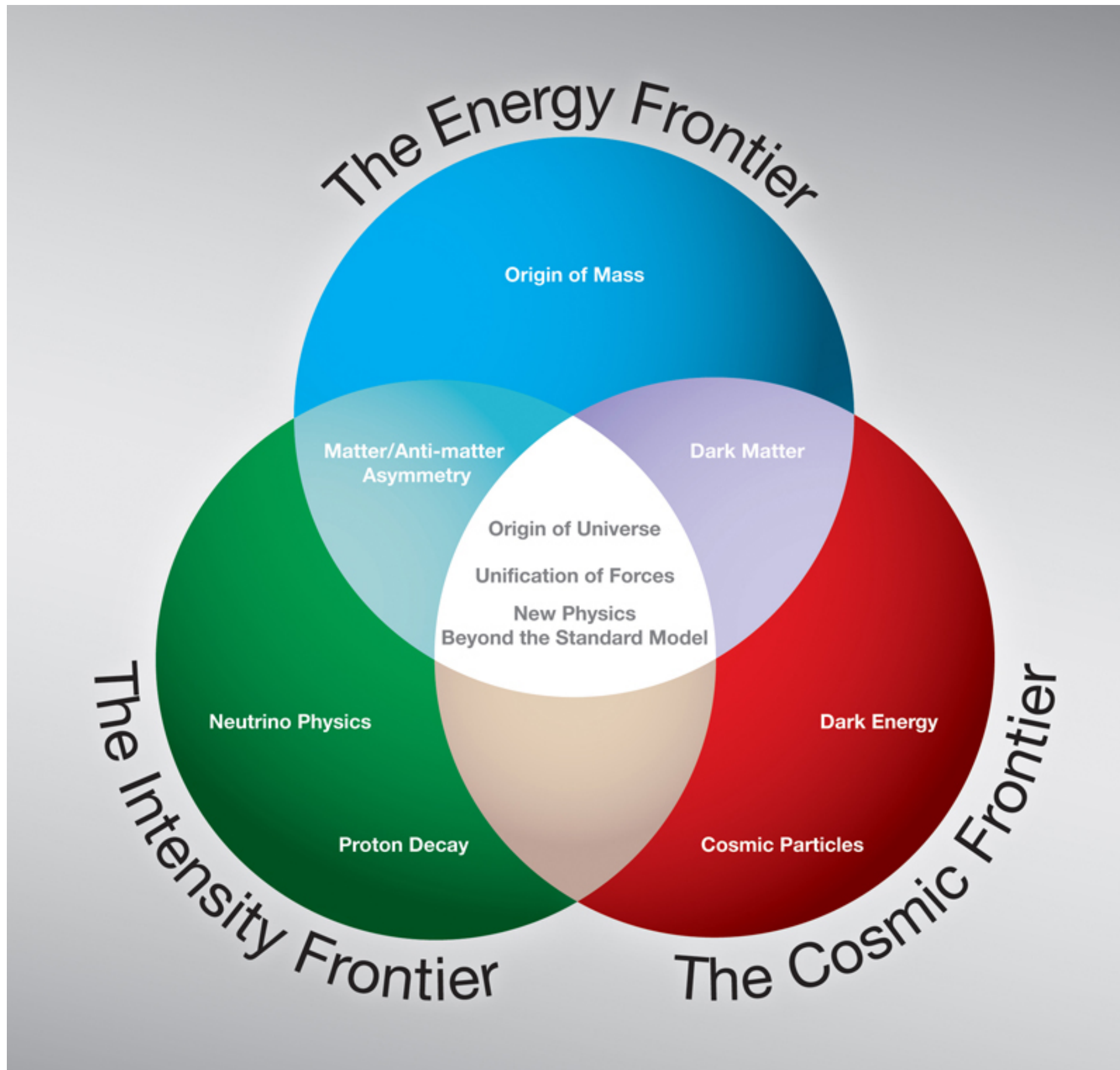


Open questions and curiosities

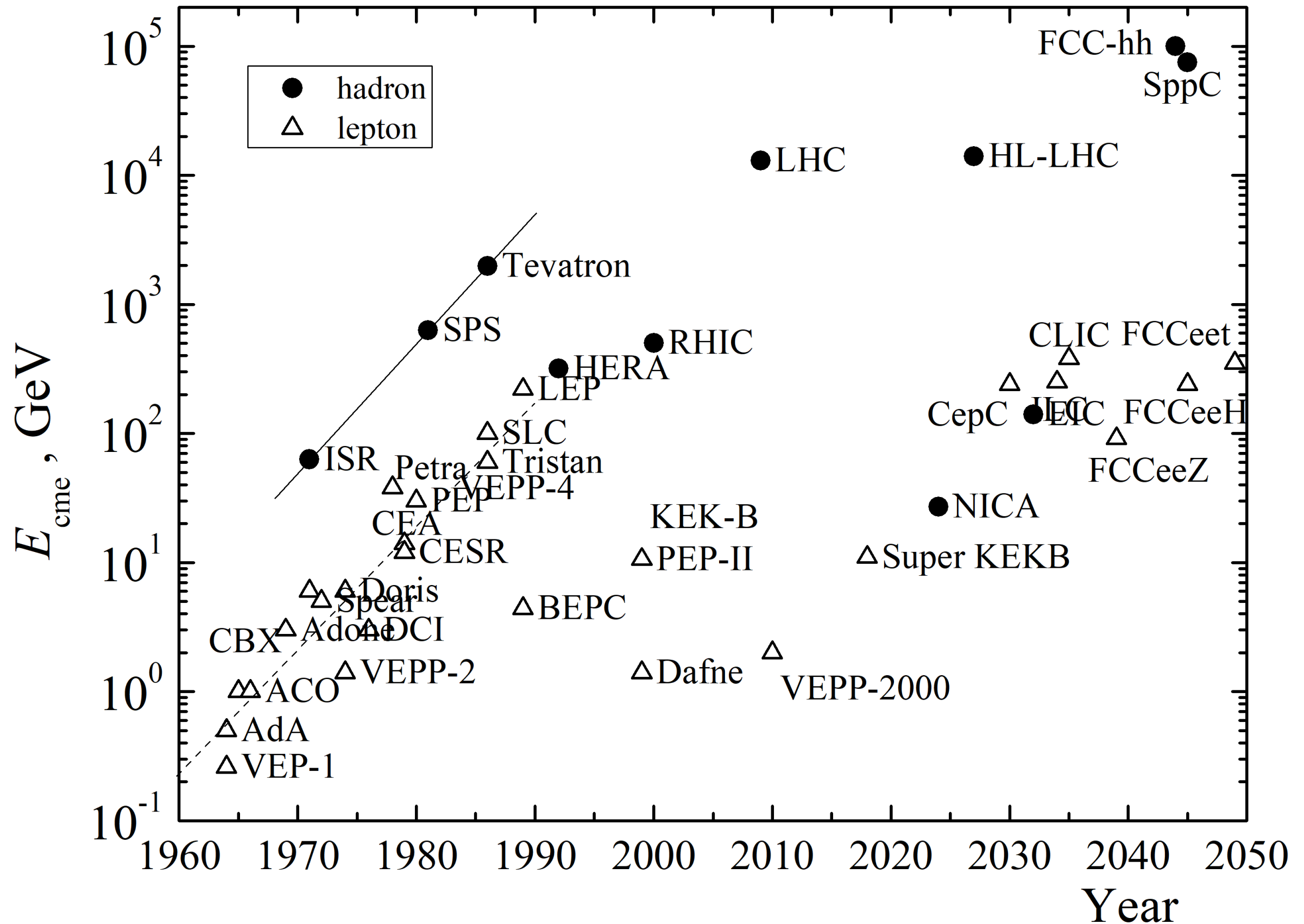
- ▷ What is the origin of mass?
- ▷ Have we found *the* Higgs boson?
- ▷ What is the origin of mass hierarchy?
- ▷ Where is all the anti-matter in our Universe?
- ▷ What is Dark Matter?



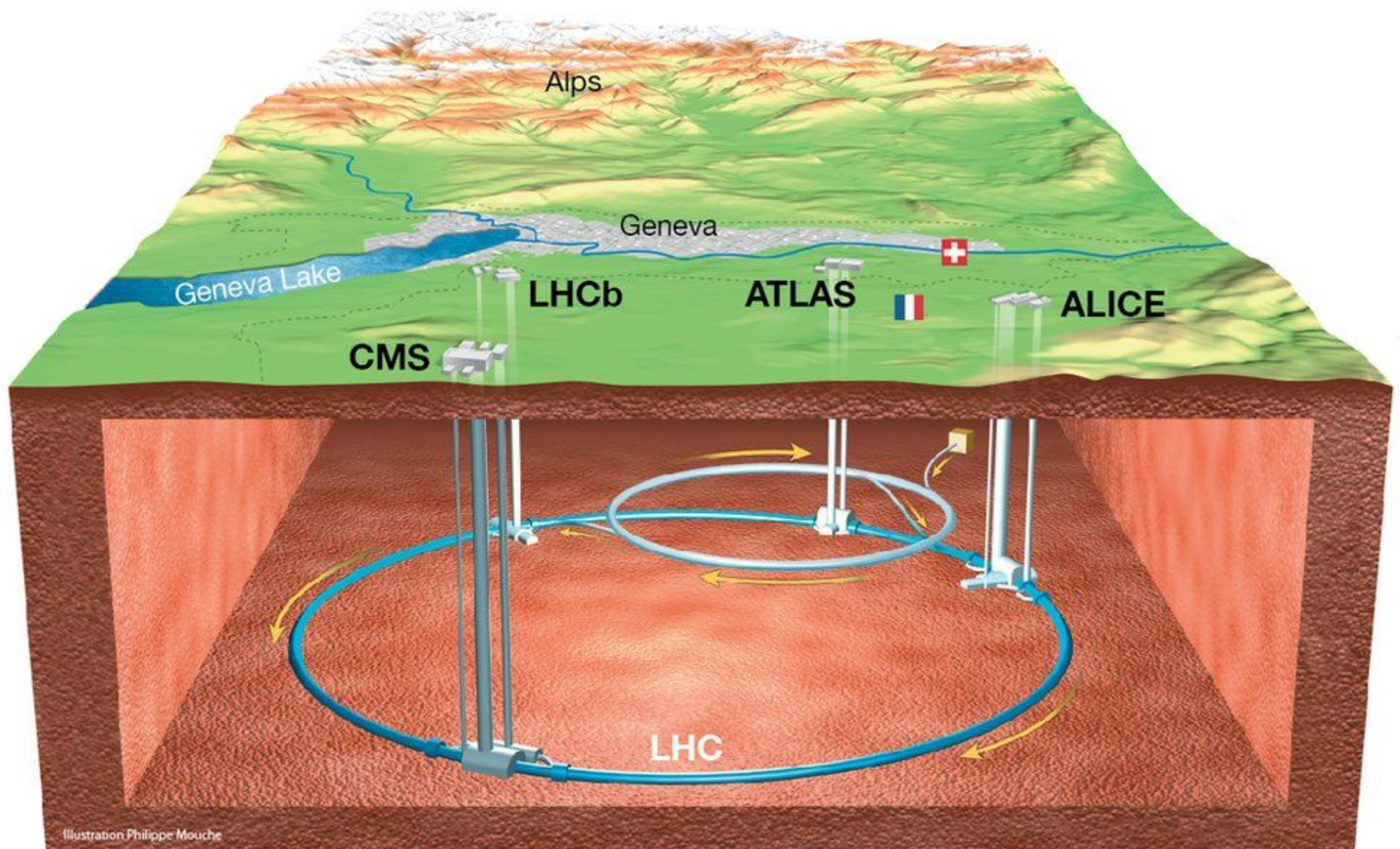
Multi-prong Approach



Evolution of Particle Colliders



Large Hadron Collider at CERN



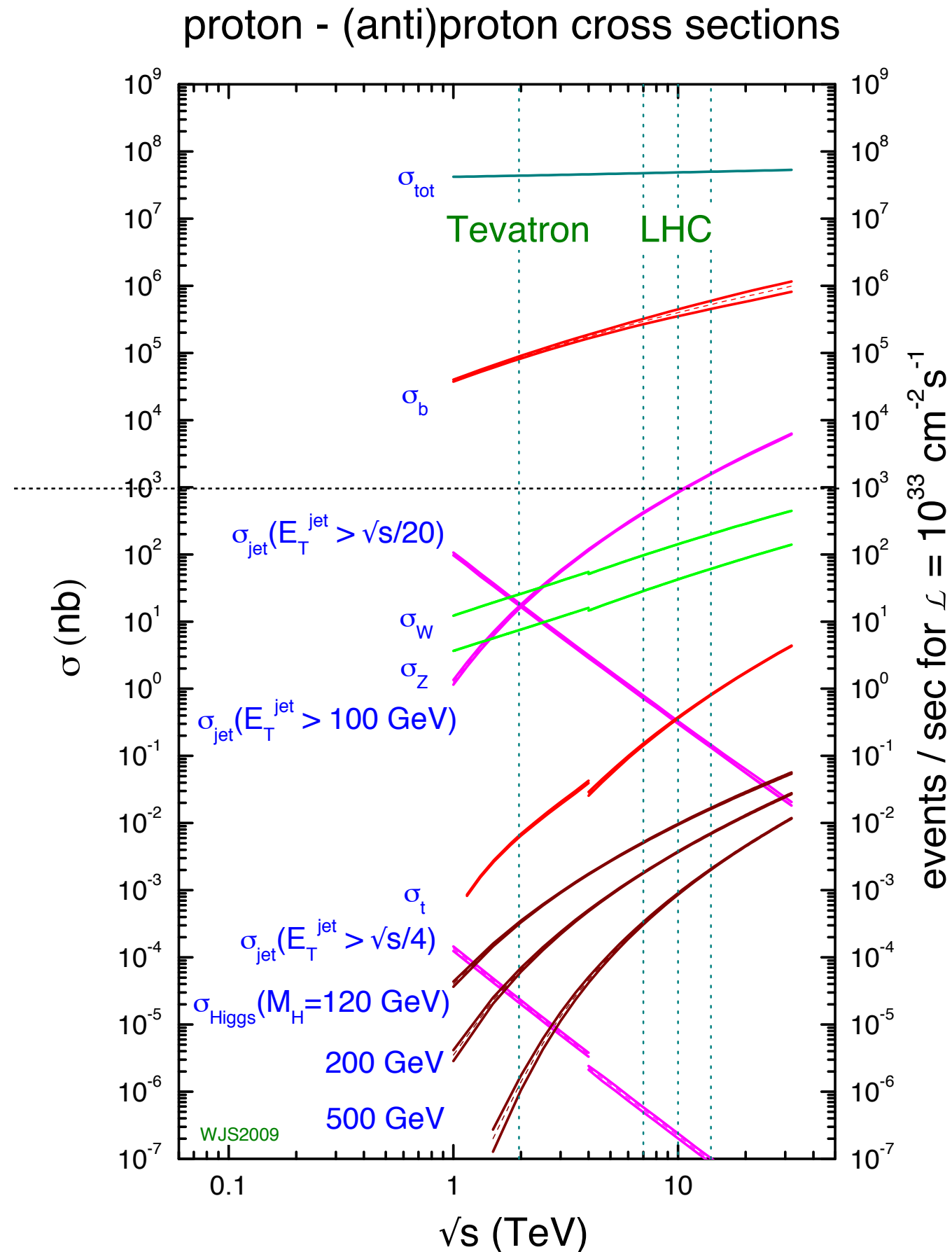
Four Experiments at LHC



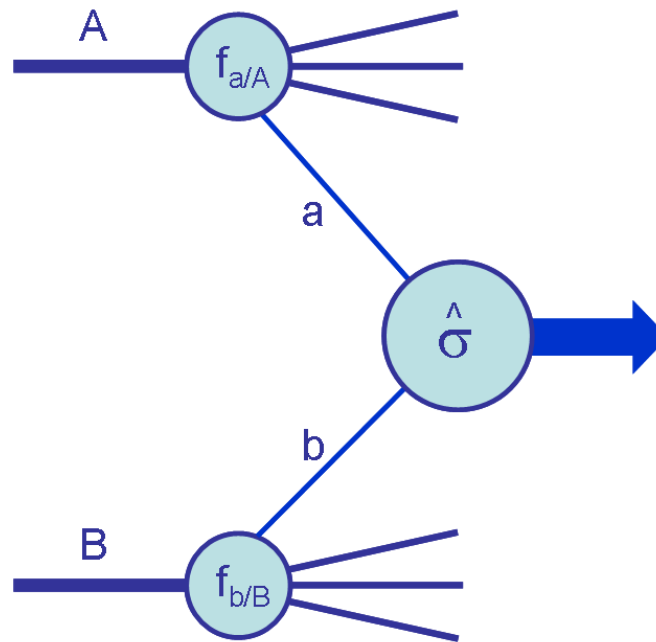
Four Experiments at LHC



Physics Production at LHC



Proton-proton Cross Section



$$\sigma_{AB} = \sum_{a,b=q,g} \left[\hat{\sigma}_{ab}^{\text{LO}} + \alpha_S(Q^2) \hat{\sigma}_{ab}^{\text{NLO}} + \dots \right] \otimes f_{a/A}(x_a, Q^2) \otimes f_{b/B}(x_b, Q^2)$$

$$\sigma_X = \sum_{a,b} \int_0^1 dx_a dx_b f(x_a, \text{flav}_a, Q^2) f(x_b, \text{flav}_b, Q^2) \cdot \sigma_{ab \rightarrow X}(x_a, x_b, Q^2)$$

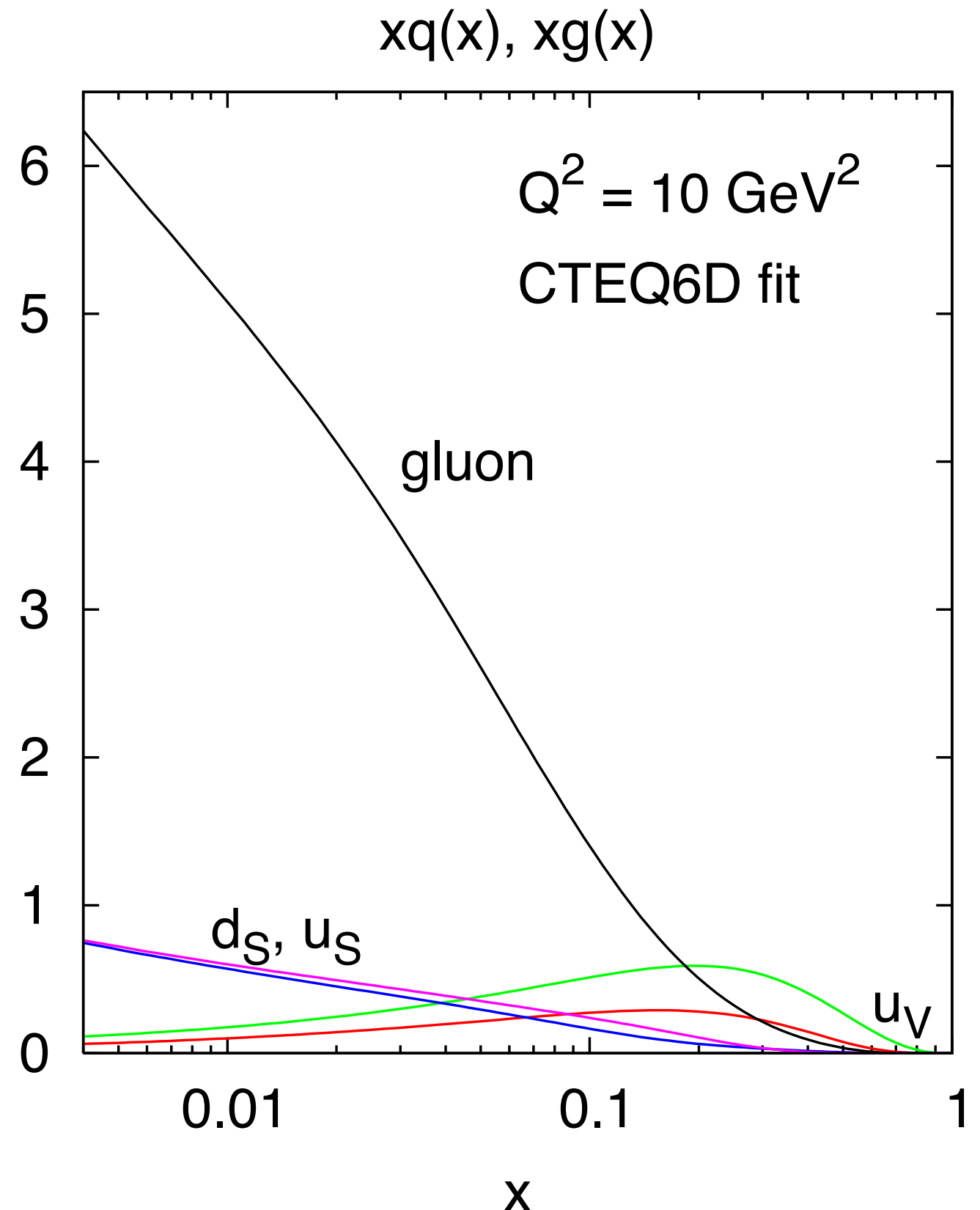
Sum over initial partonic states a,b

Parton Density Function

hard scattering cross-section

Gluon Density Function

- ▷ Gluons dominate by far at low x
- ▷ LHC dominated mainly by gluon-gluon fusion (hard scattering of 2 gluons) at low x
- ▷ Different experimental signatures for qq , q -anti- q and qg hard scattering
 - also very different cross sections



Pictorial Representation of p-p Collision

Scetch of a proton–proton collision
at high energies

Initial state parton shower

Signal process = production of jets

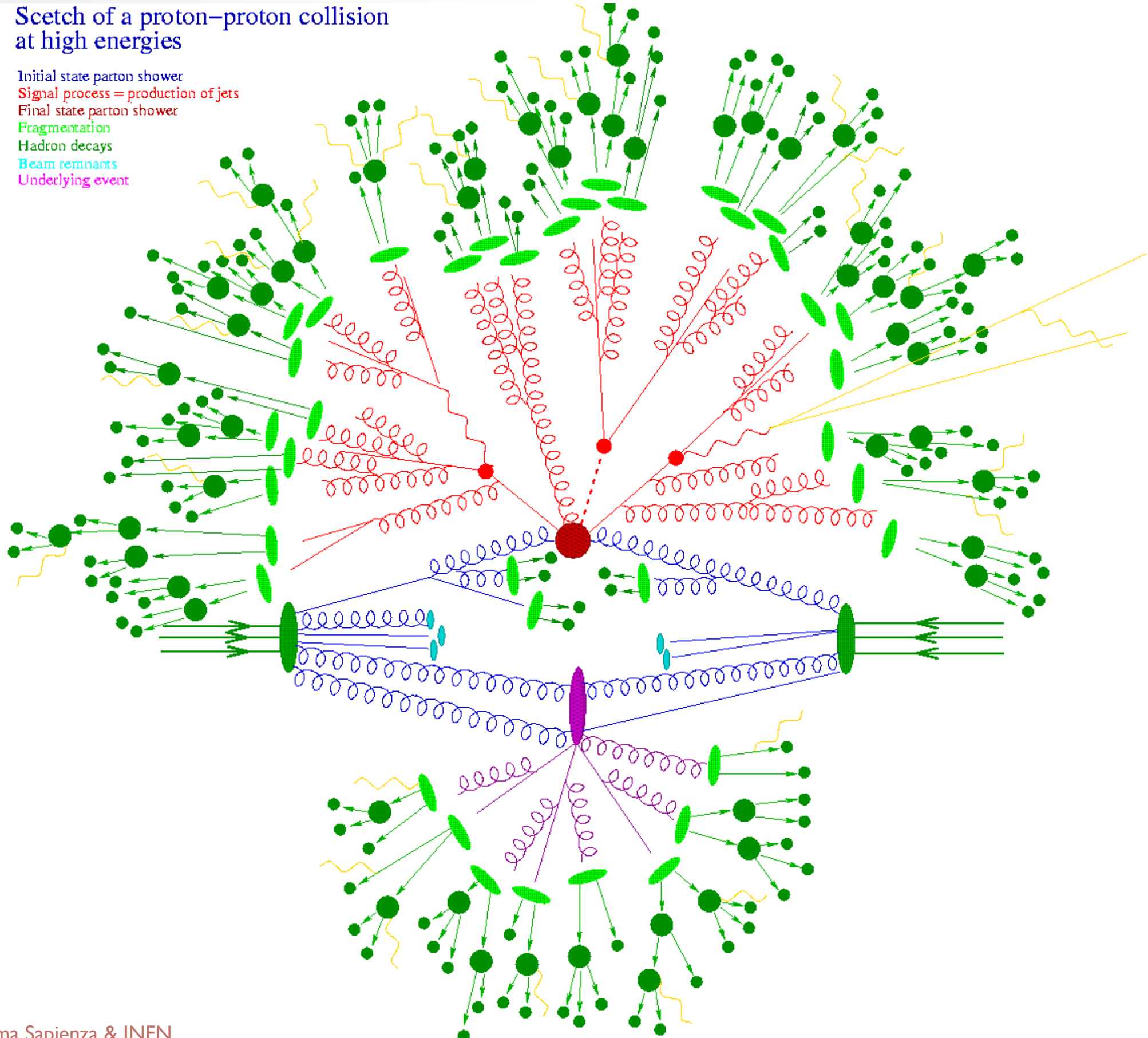
Final state parton shower

Fragmentation

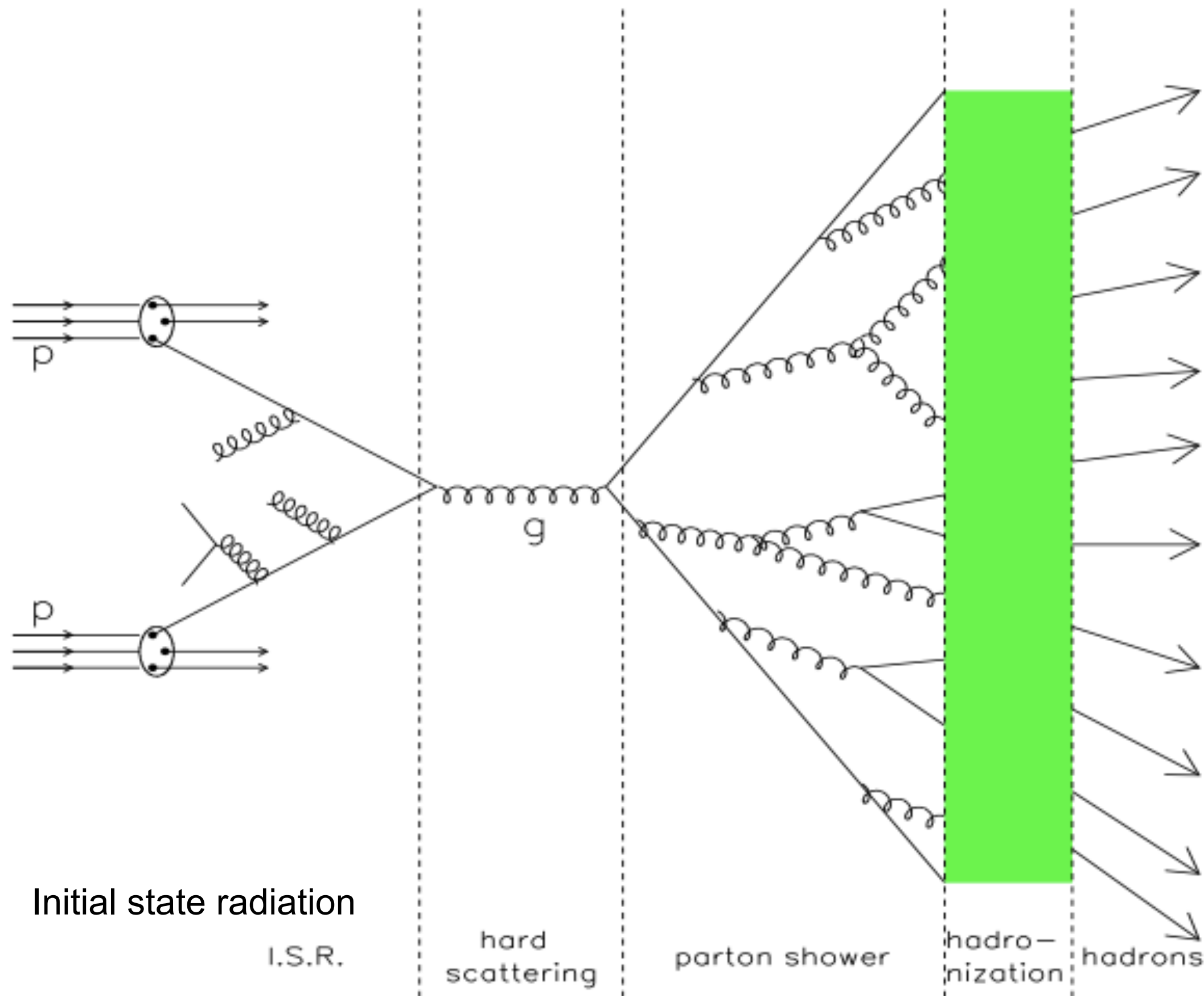
Hadron decays

Beam remnants

Underlying event



Hadron Hadron Collisions



Detectors

Pile-Up

▷ High probability of more than 1 proton-pair interaction with

- smaller beam section
- higher number of protons per bunch

▷ Typically one interesting hard-scattering of interest

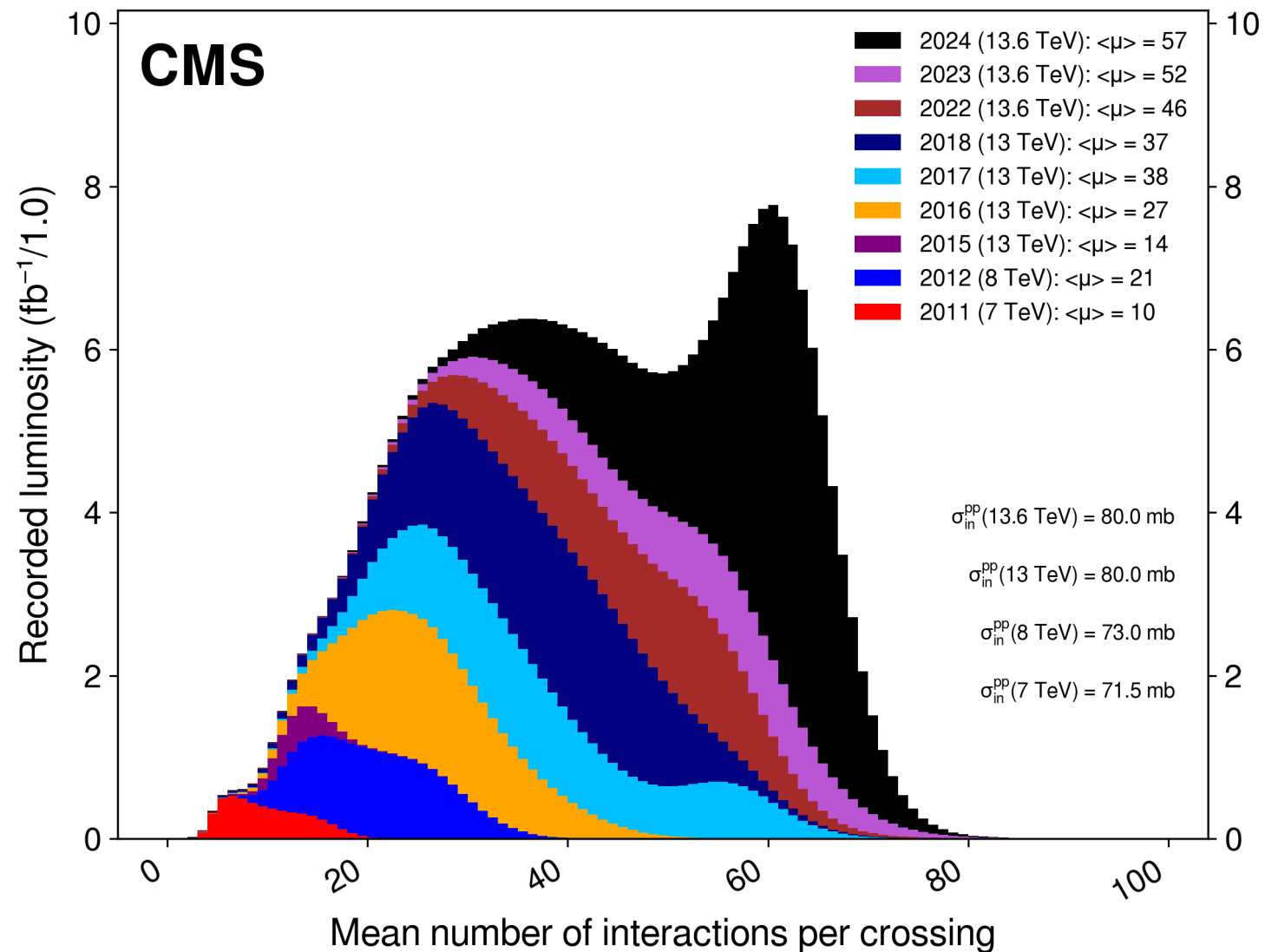
- the “event” causing the trigger

▷ Additional collisions in same bunch crossing increase average energy density in detectors

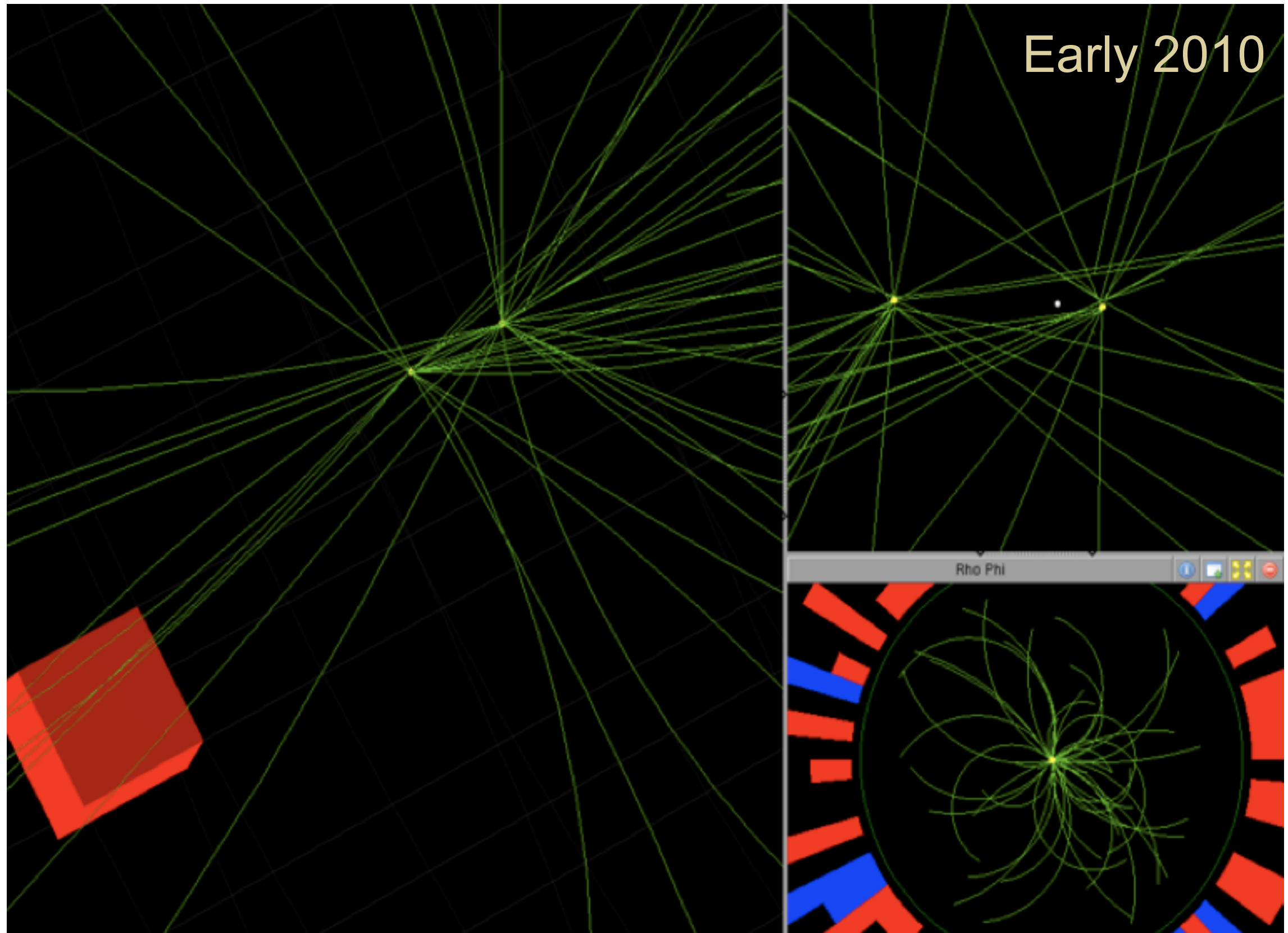
- higher occupancy
- distortion or mis-measurement of energy in calorimeters
- Potential inefficiency in track reconstruction in very busy events

▷ Much larger event processing time

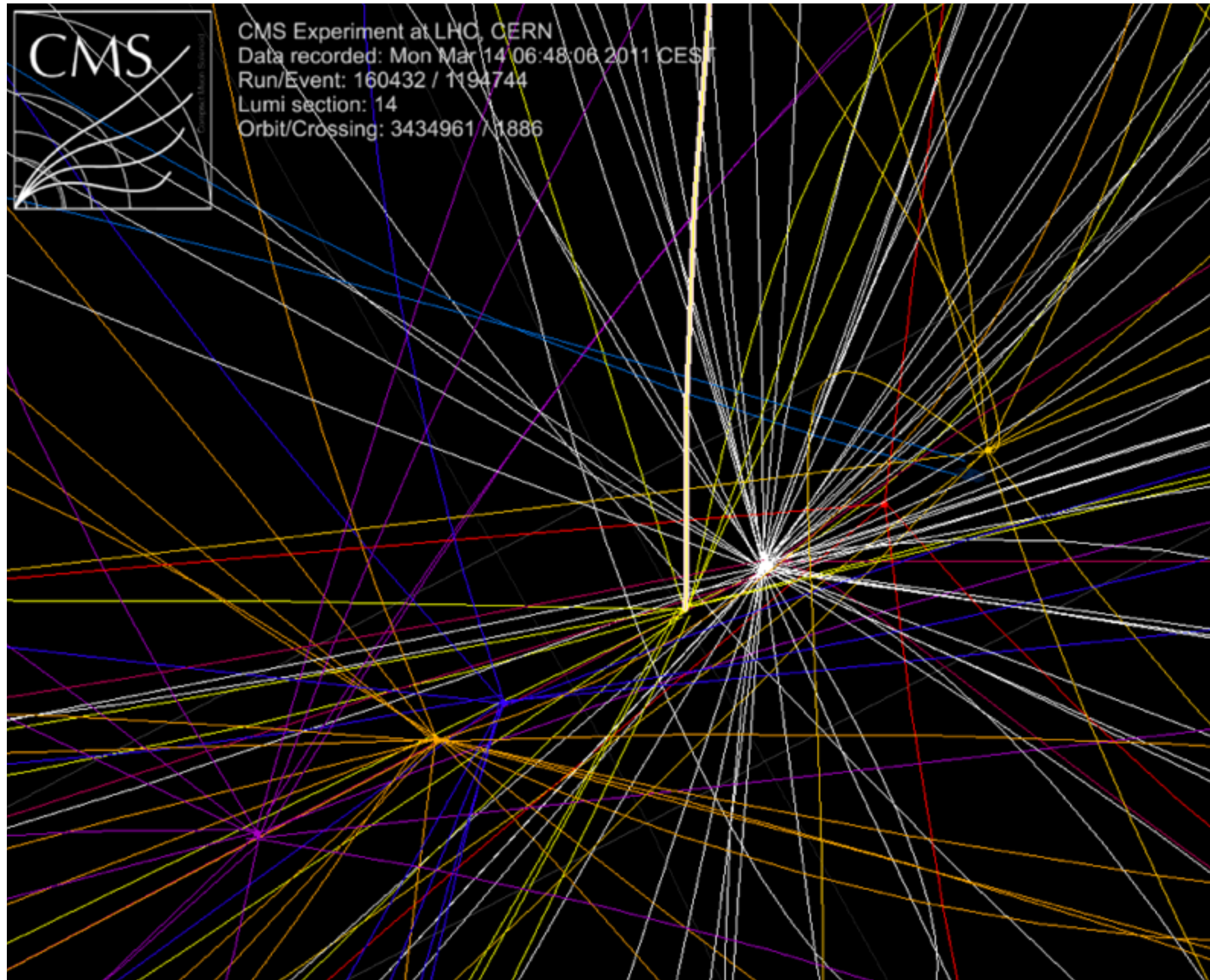
- Increasing computing power to keep up with data



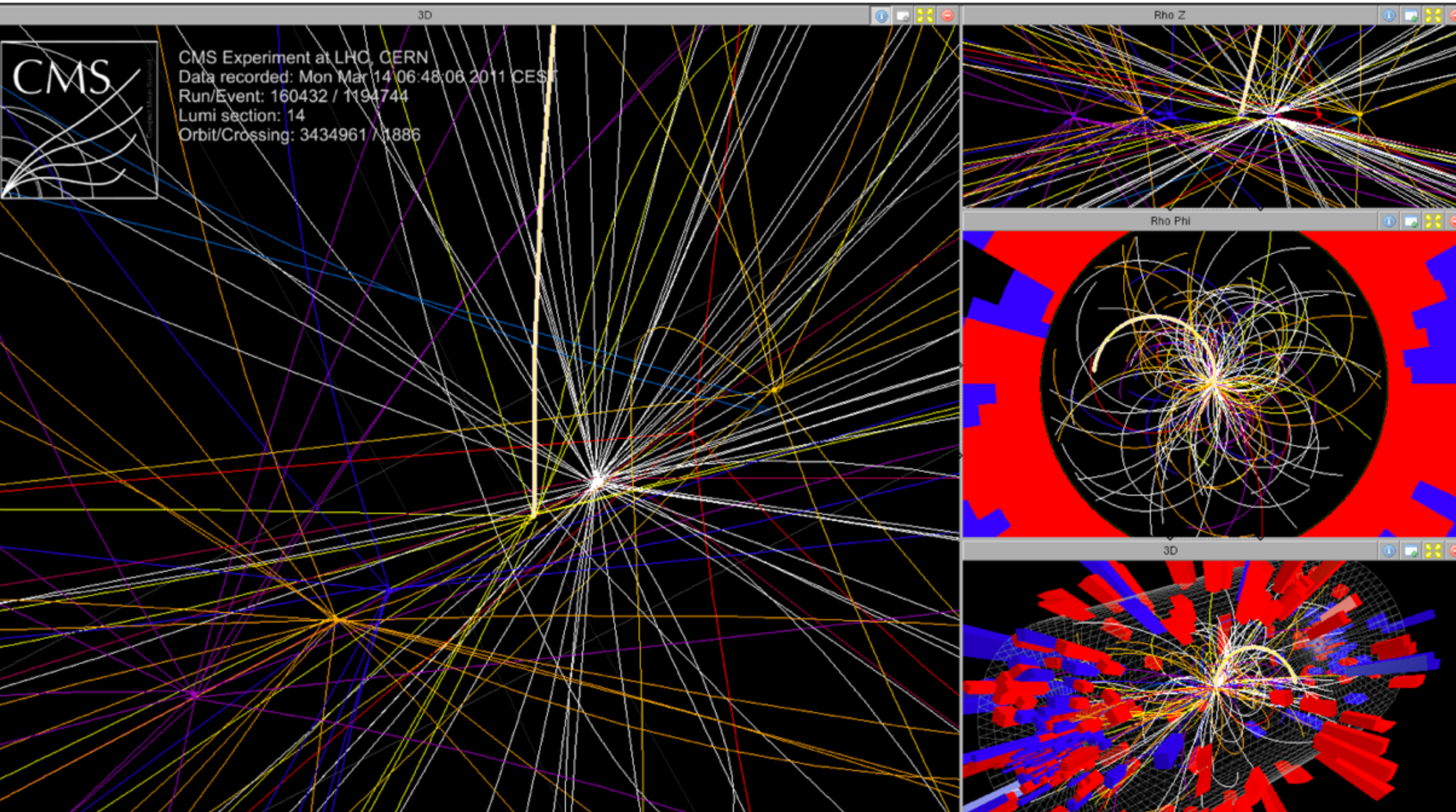
Pile-Up at Low Luminosity



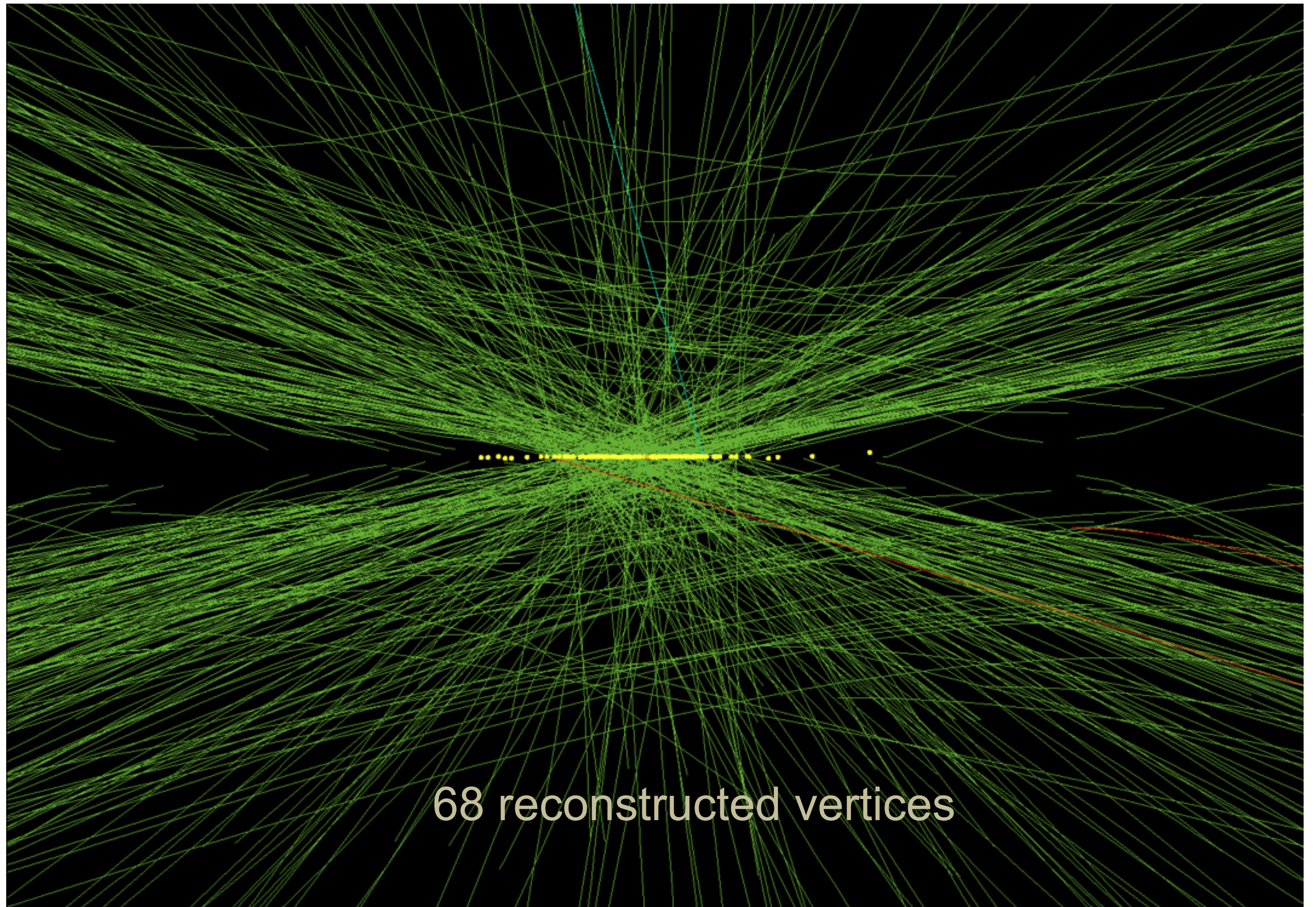
Pile-Up Early 2011



Pile-Up Early 2011



High Pile-Up Event in 2012

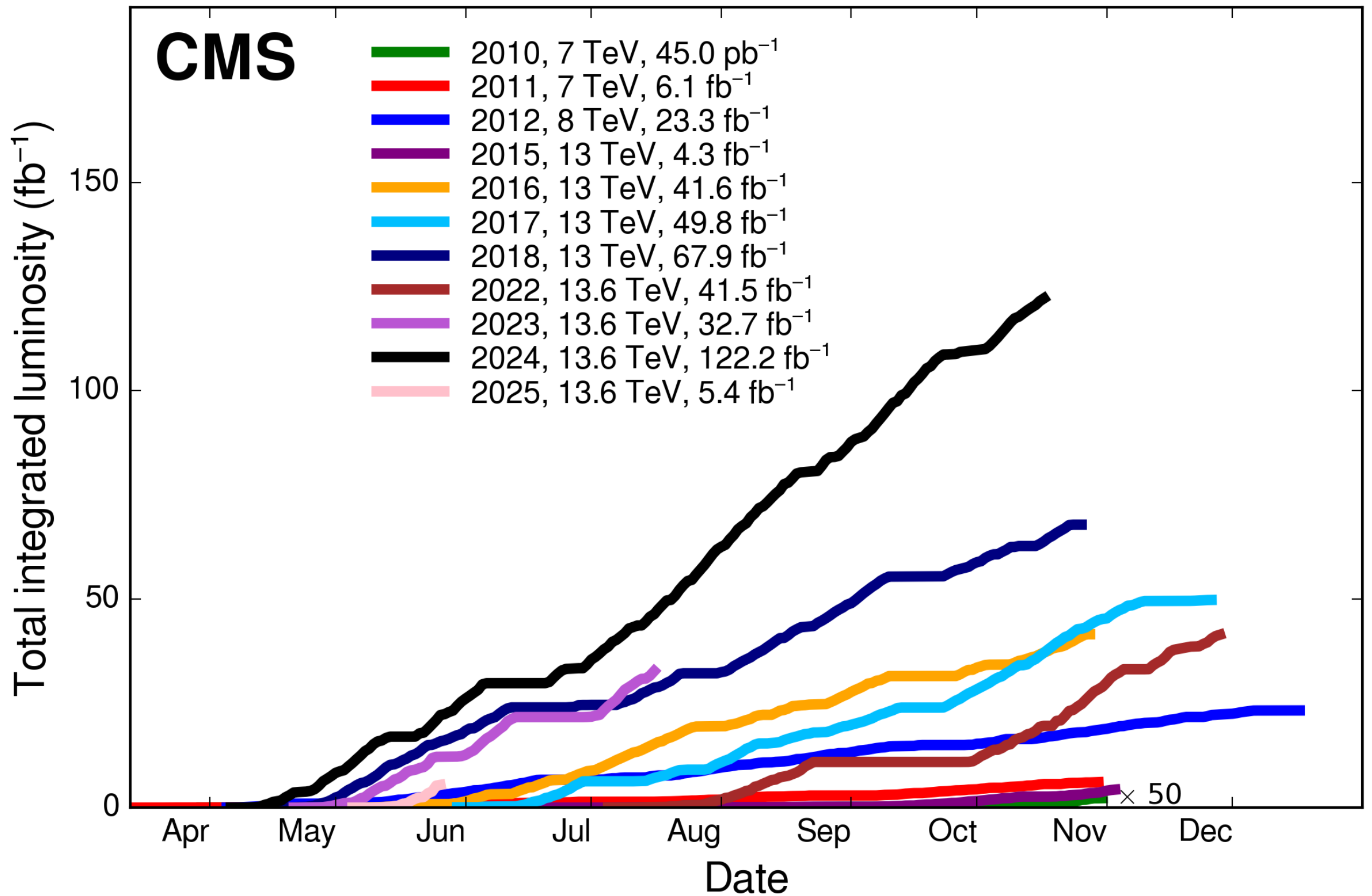


$Z \rightarrow \mu\mu$ event from 2012 data with 25 reconstructed vertices

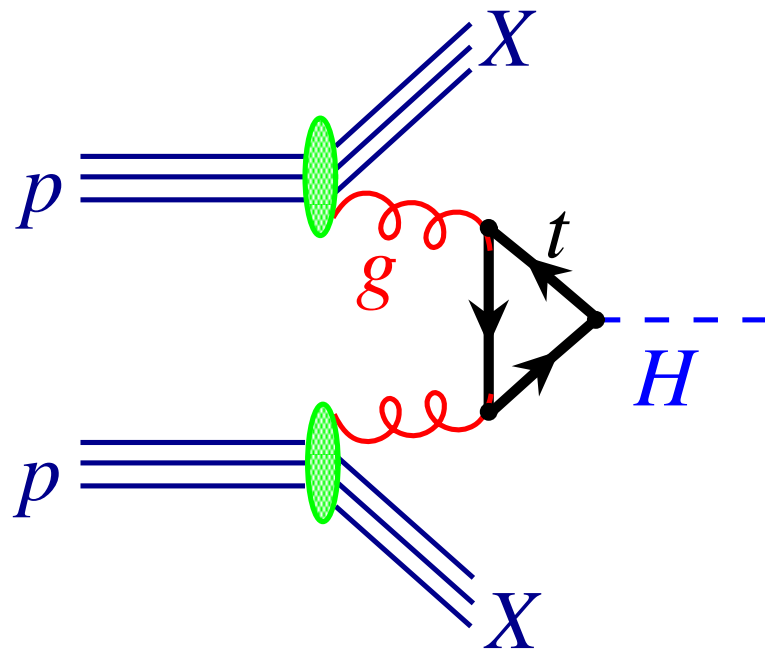
$Z \rightarrow \mu\mu$



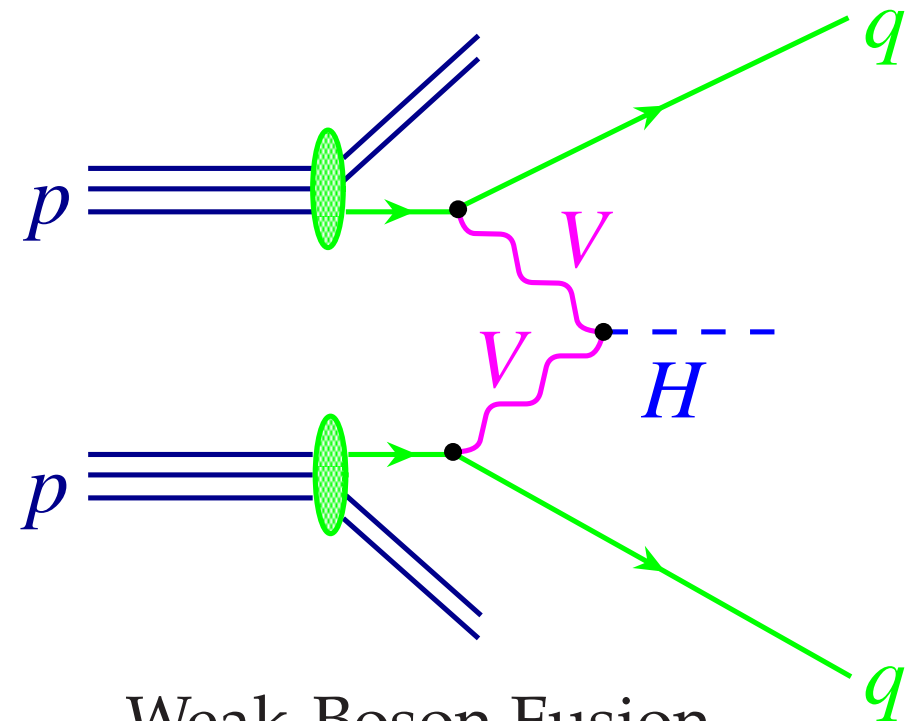
Integrated luminosity



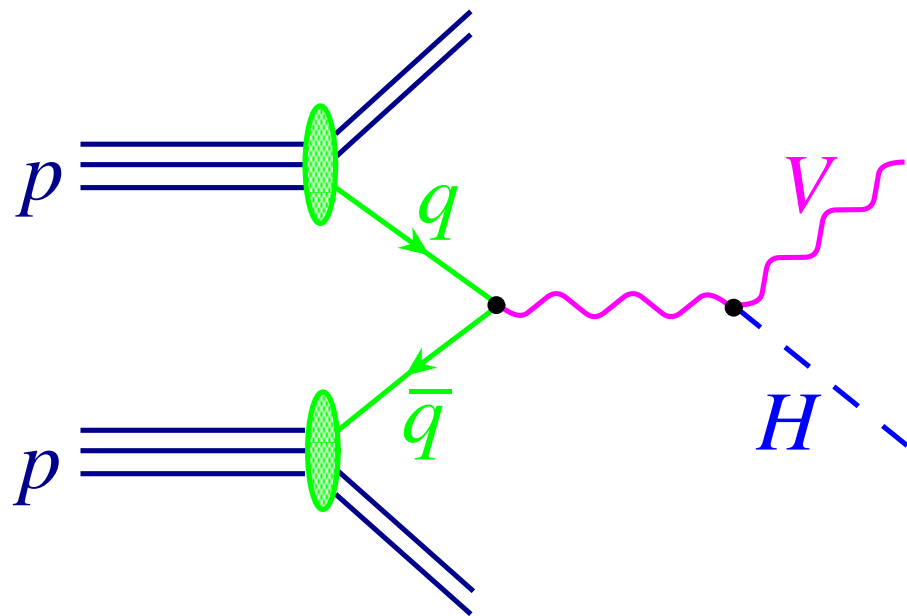
Higgs Production at Hadron Colliders



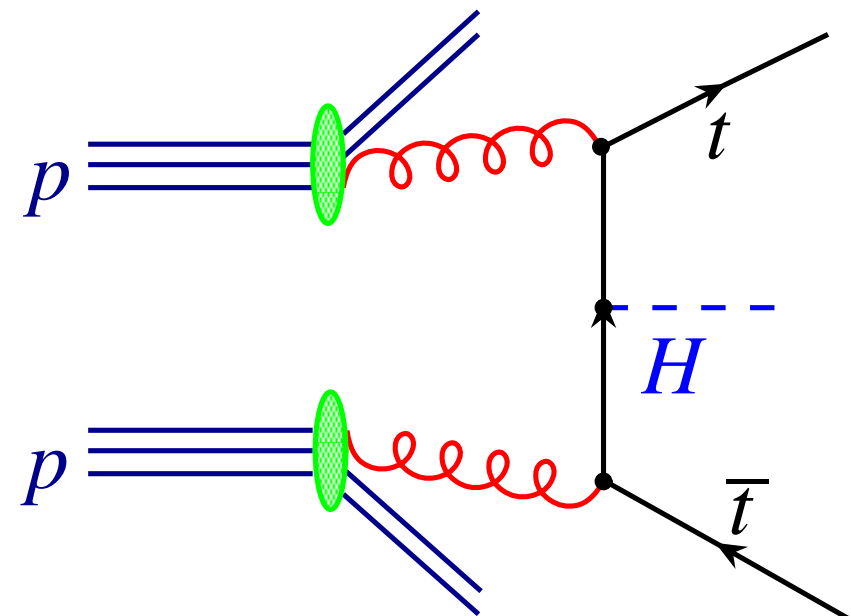
Gluon fusion



Weak-Boson Fusion

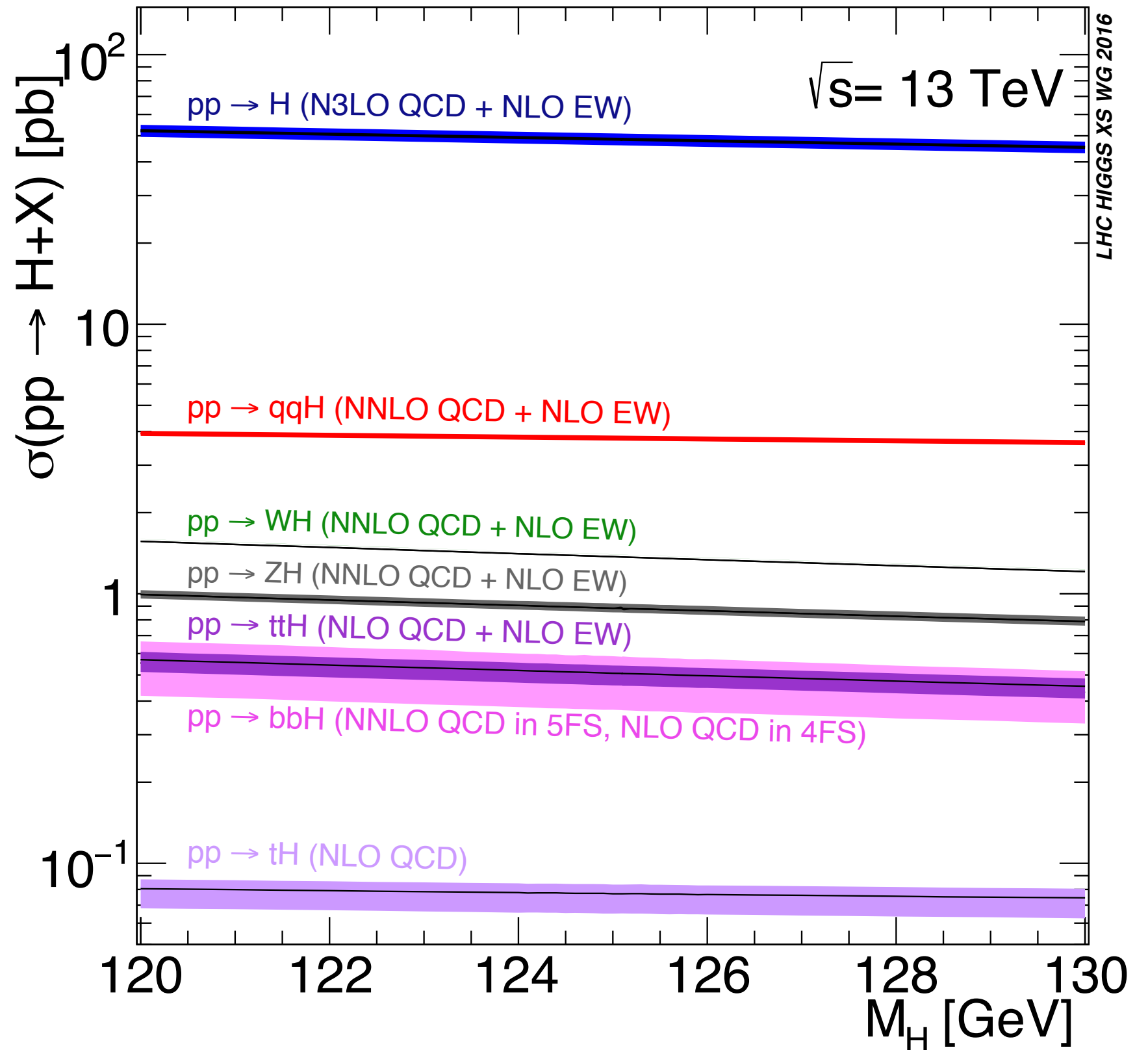
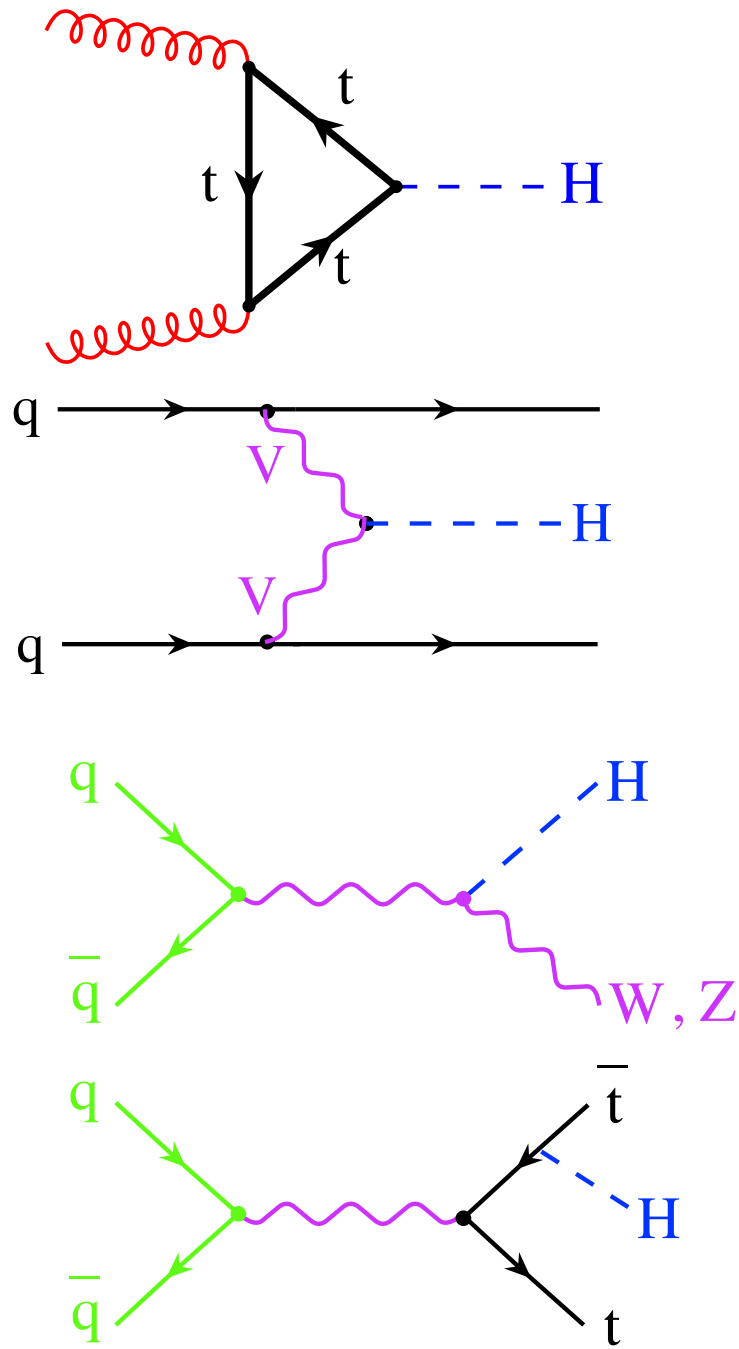


Higgs Strahlung

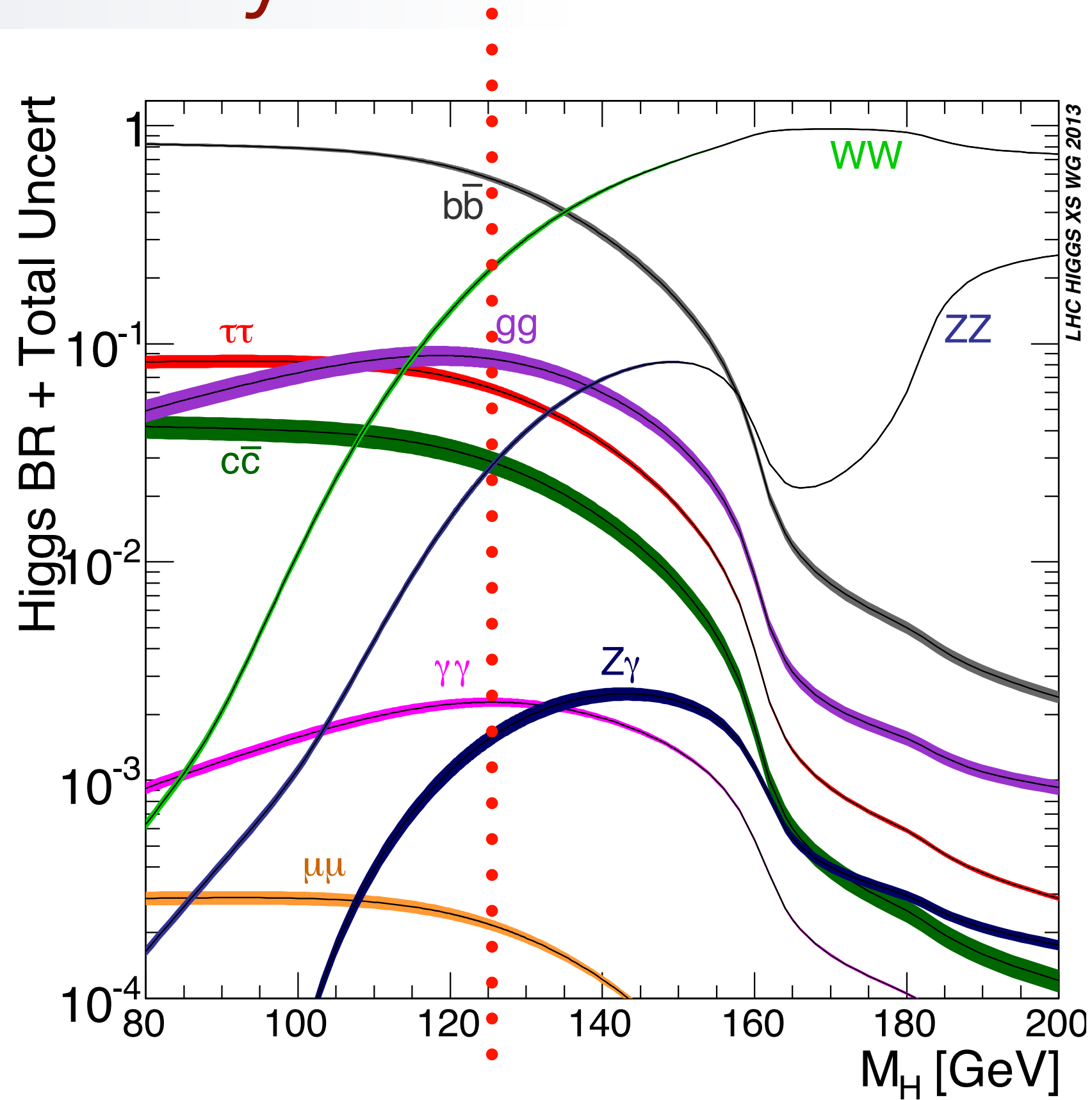


$t\bar{t}H$

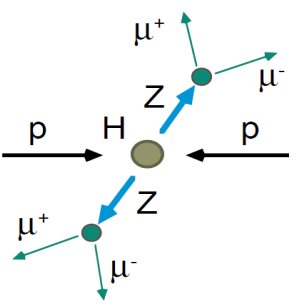
Higgs Production Cross Section



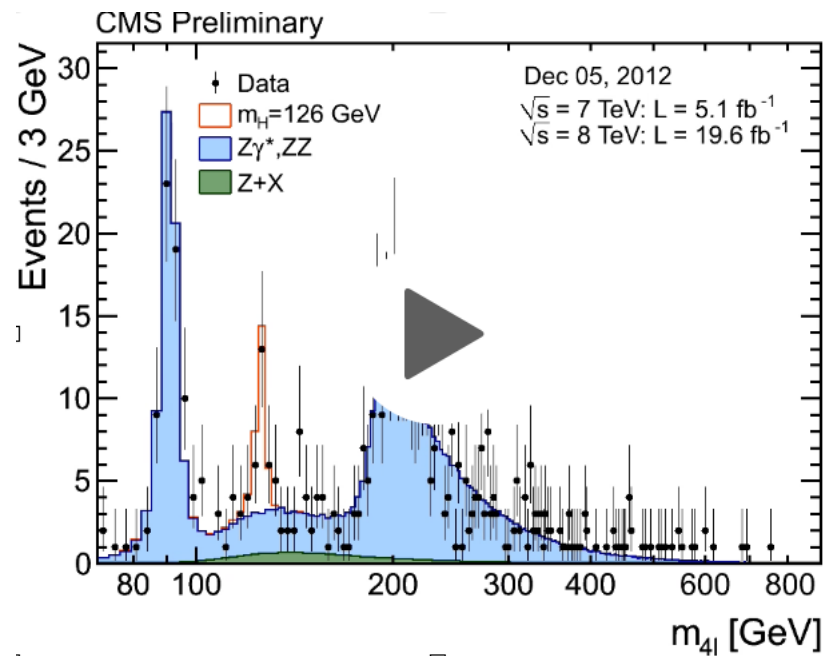
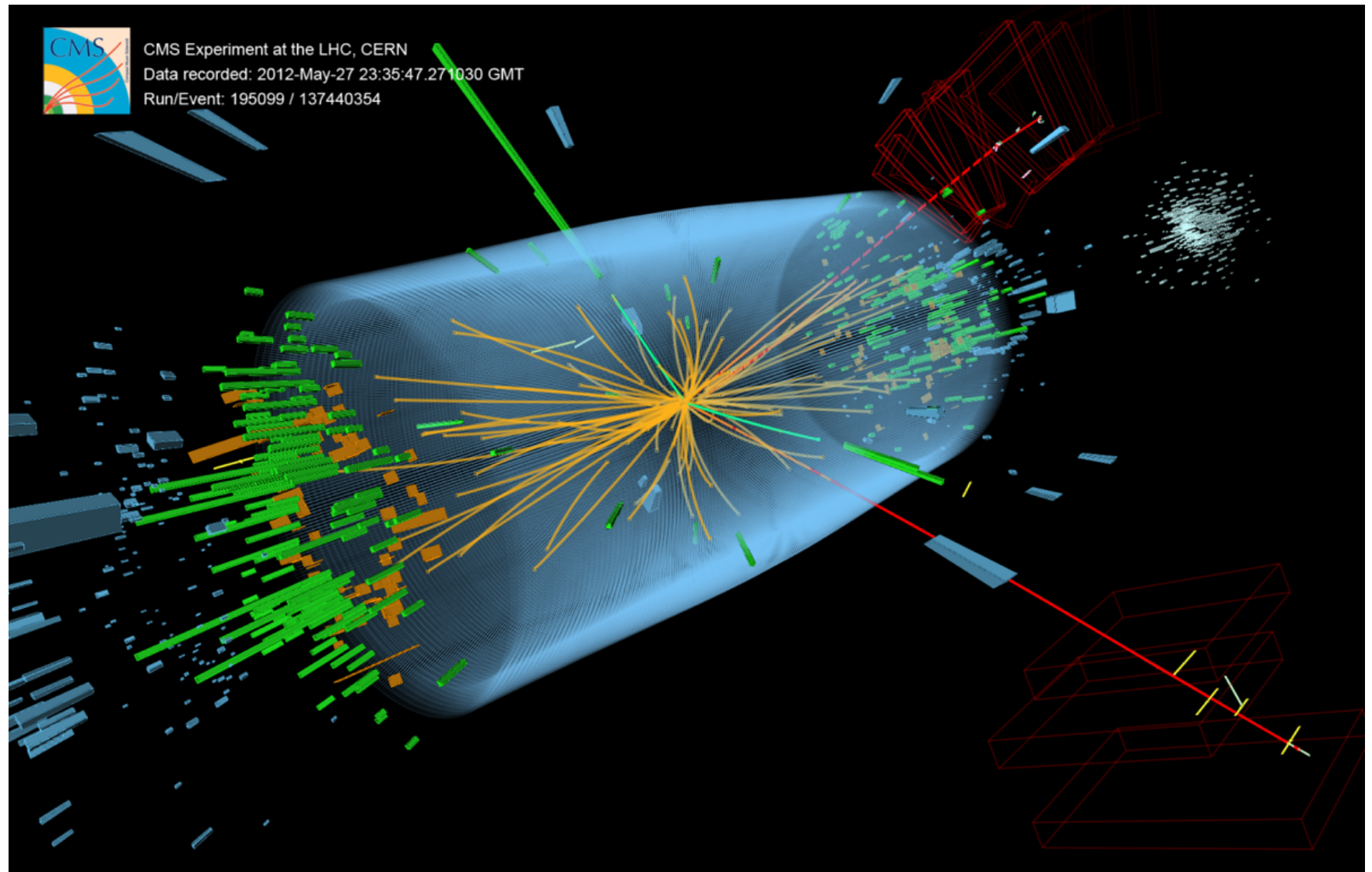
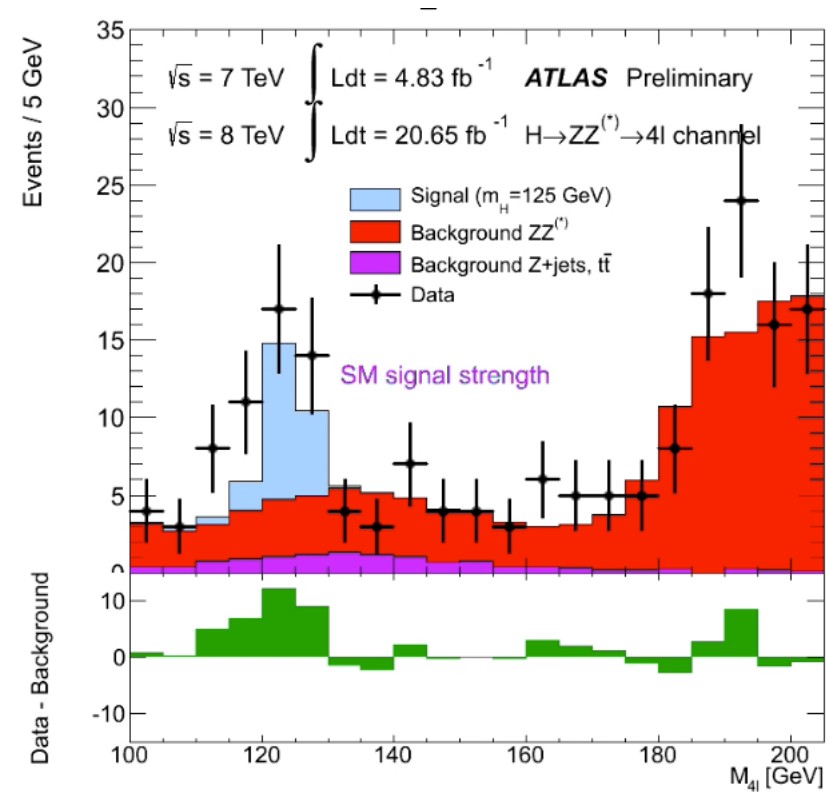
Higgs Decay



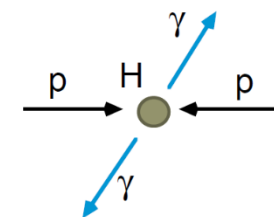
Higgs decay to ZZ (leptons)



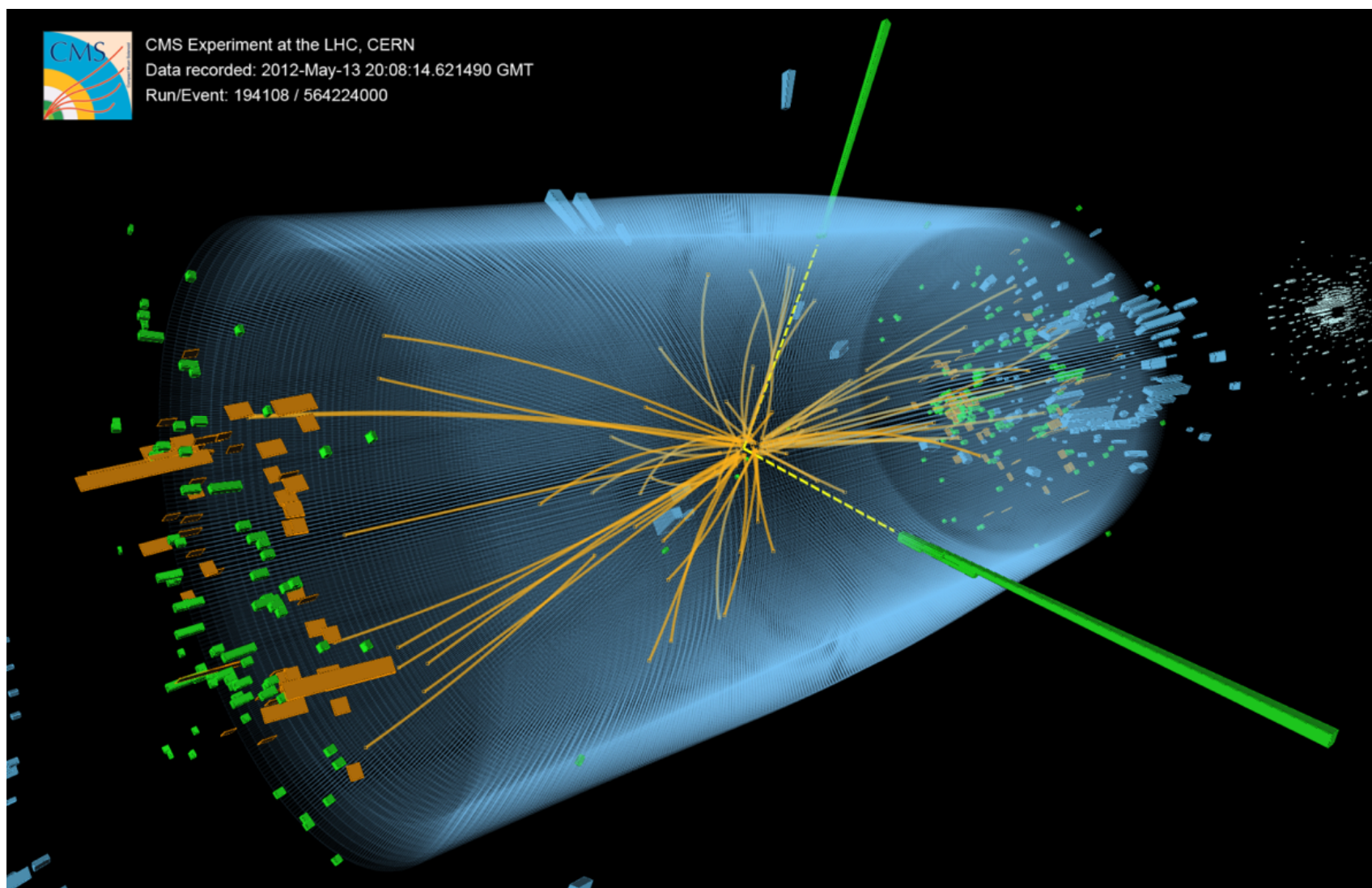
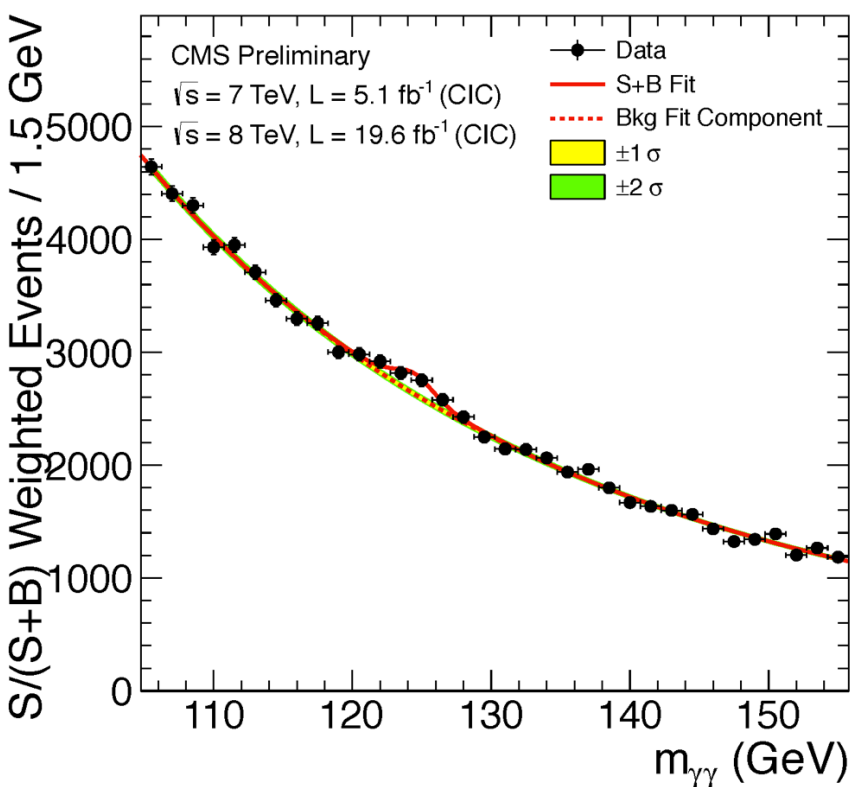
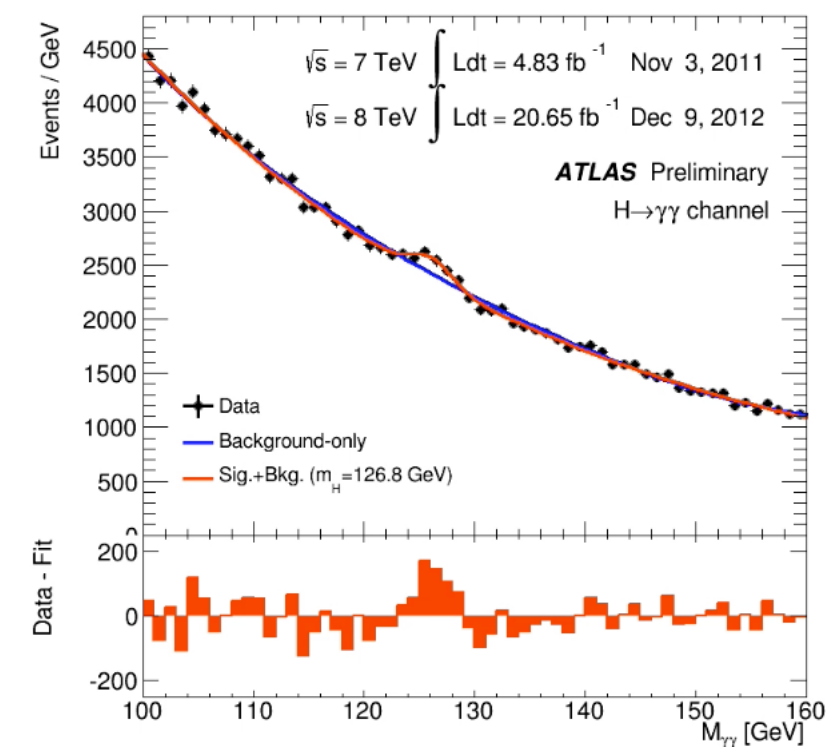
$$mass = \sqrt{\vec{p}_1^2 + \vec{p}_2^2 + \vec{p}_3^2 + \vec{p}_4^2}$$



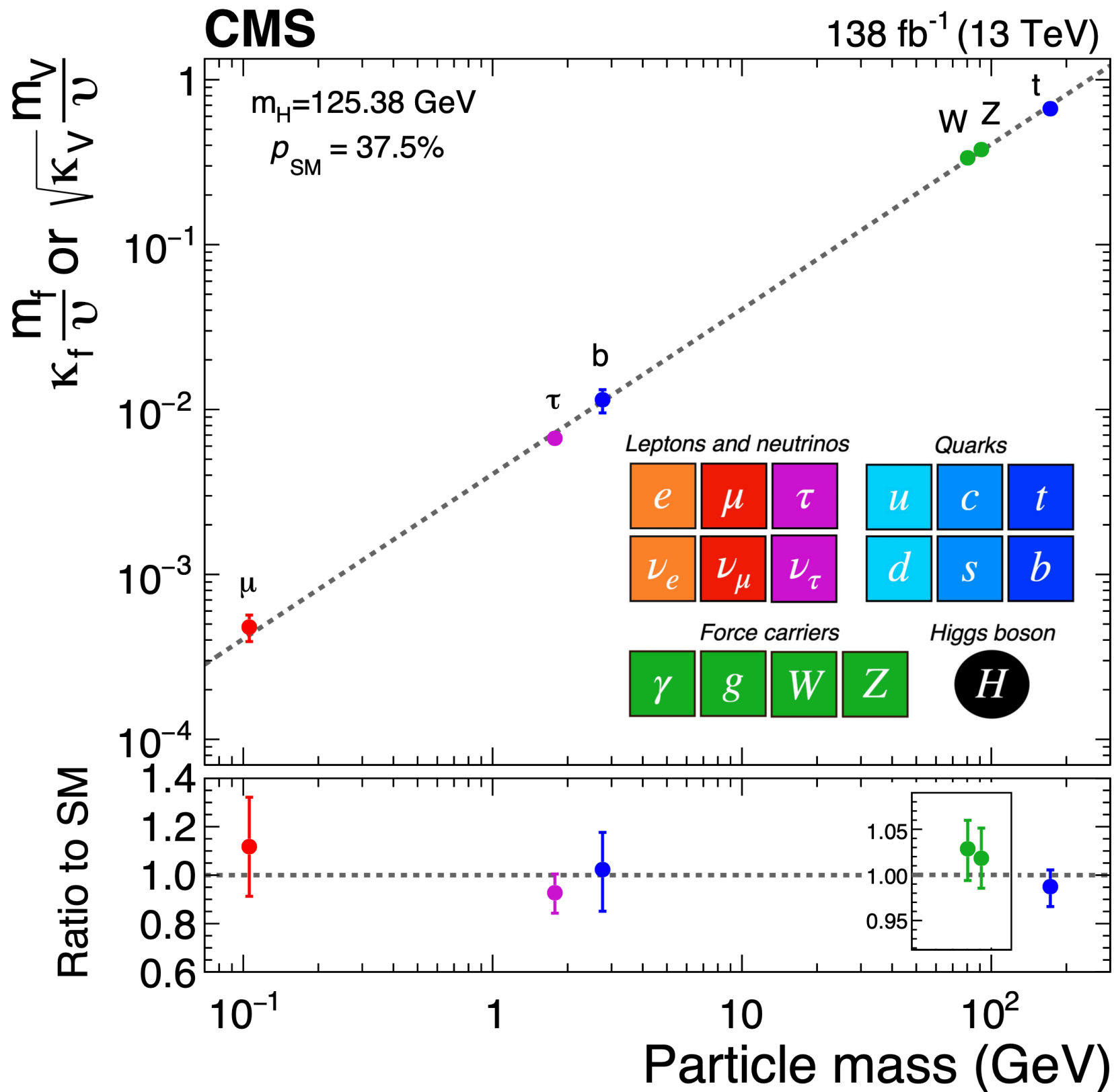
Higgs decay to Photons



$$mass = \sqrt{2E_{\gamma_1}E_{\gamma_2}(1 - \cos \theta_{12})}$$



Higgs couplings today



Energy Frontier after Higgs Discovery

- ▷ Intense scrutiny of Higgs and Yukawa sector

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}D\psi$$

Precision Electroweak and
QCD

$$+|D_{\mu}\phi|^2 - V(H)$$

Higgs properties
Higgs self interaction

$$+Y_{ij}\psi_i\psi_j\phi + \text{h.c.}$$

Higgs coupling to bosons and fermions
CKM matrix and CP Violation

- ▷ While keeping a wide open eye on new phenomena

$$+\mathcal{L}_{\text{New}}$$

New light and heavy particles
Lepton flavour universality violation
Leptoquarks
SUSY
Long-lived particles
Dark matter

Outlook

- LHC has established Yukawa interaction for third generation fermions
- Extensive program searching for new phenomena beyond the Standard Model
- Precision measurement of top and electroweak processes sensitive to deviation due to new physics
- *Exploration of new territory
for the first time without solid theoretical guidance*